

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM



CURRICULUM AND SYLLABUS FOR THE COURSES OF B.TECH/B.TECH (HON)/DUAL DEGREE (B.TECH - MS) PROGRAMME IN COMPUTER SCIENCE AND ENGINEERING WITH SPECIALIZATION IN CYBER SECURITY

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B.Tech/B.Tech (Hon)/ B.Tech - MS General Course Structure CSE with Specialization in Cyber Security

Semester -I						Semester -II					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA111	Discrete Mathematics	3	1	0	4	IMA121	Calculus and Linear Algebra	3	1	0	4
ICS111	IT Workshop I	3	1	2	5	IEC121	Digital Design and Electric Circuits	3	1	2	5
ICS112	Computer Programming	3	1	2	5	ICS121	Data Structures I	3	1	2	5
IEC111	Electronic Circuits	3	1	2	5	ICS122	Computer Organization	3	1	0	4
IHS111	Communication Skills	3	0	0	3	ICS123	IT Workshop II	2	1	2	4
IHS112	Foreign Language	1	0	0	1	IHS121	Personality Development	1	0	0	1
IPT111	Physical Training I	0	0	2	0						
Total		16	4	8	23	Total		15	5	6	23
Cumulative Credits at the End of First Year: 46											
Semester-III						Semester -IV					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA211	Probability, Statistics and Random Processes	3	1	0	4	IMA221	Differential Equations and Transforms	3	1	0	4
ICS211	Design and Analysis of Algorithms	3	1	0	4	ICS221	Operating Systems (2024 onwards)	3	1	0	4
ICS212	Theory of Computation (2024 onwards)	3	1	0	4	ICS226	Secure Software Engineering	3	0	2	4
ICS213	Database Management Systems	2	1	2	4	ICS223	Compiler Design	3	0	2	4
ICS214	IT Workshop III	2	1	2	4	ICS224	Computer Networks	3	0	2	4
ISC212	Quantum Computing and Security	2	0	0	2	ICS225	Data Structures III	1	0	2	2
ICS215	Data Structures II	1	0	2	2	IHS221	Fundamentals of Economics	1	0	0	1
IPT211	Physical Training II	0	0	2	0	IHS223	Risk Management	1	0	0	1
Total		16	5	8	24	Total		18	2	8	24
Cumulative Credits at the End of Second Year: 94											
Semester-V						Semester -VI					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CBEXXX	Bouquet Core I	3	0	0	3	CBEXXX	Bouquet Core III	3	0	0	3
CBEXXX	Bouquet Core II	3	0	2	4	CBEXXX	Bouquet Core IV	3	0	2	4
CBSXXX	CB Elective I	3	0	2	4	CBEXXX	Bouquet Core V	3	0	2	4
IEGXXX	Engineering Elective	3	0	2	4	CBSXXX	CB Elective III	3	0	2	4
IMAXXX	Mathematics Elective	3	0	0	3	ISCXXX	Science Elective	3	0	0	3
IHSXXX	Humanities Elective	3	0	0	3	IOEXXX	Institute Open Elective I	3	0	0	3
CBEXXX	CB Elective II	1	0	0	1	IHP325	Honours Project Phase I (Optional)				
Total		19	0	6	22	Total		18	0	6	21
Cumulative Credits at the End of Third Year: 137											
Semester -VII						Semester -VIII					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CBSXXX	CB Elective IV	3	0	2	4	CBSXXX	CB Elective V	3	0	0	3
CBEXXX	Bouquet Core VI	3	0	2	4	CBSXXX	CB Elective VI /Industrial Training	3	0	2	4
CBEXXX	Bouquet Core VII	3	0	2	4	IOEXXX	Institute Open Elective III	3	0	0	3
IOEXXX	Institute Open Elective II	3	0	0	3	IBP423	B. Tech Project Phase II	6	0	0	6
IBP413	B. Tech Project Phase I	6	0	0	6	IHP425	Honours Project Phase II (Optional)				
CBSXXX	Research Course (Optional)					CBSXXX	Research Course (Optional)				
Total		18	0	6	21	Total		15	0	2	16
Cumulative Credits at the End of Fourth Year: 174 (B.Tech); 174+12=186 (B.Tech (Hon)); 174+20=194 (B.Tech-MS)											
Semester -IX						Semester -X					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IRP511	Research Project	12	0	0	12	IRP521	Research Project	12	0	0	12
Total		12	0	0	12	Total		12	0	0	12
Cumulative Credits at the End of Fifth Year: 218 (B.Tech-MS)											

B.Tech/B.Tech (Hon)/ B.Tech - MS Detailed Course Structure CSE with Specialization in Cyber Security

Semester -I						Semester -II					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA111	Discrete Mathematics	3	1	0	4	IMA121	Calculus and Linear Algebra	3	1	0	4
ICS111	IT Workshop I	3	1	2	5	IEC121	Digital Design and Electric Circuits	3	1	2	5
ICS112	Computer Programming	3	1	2	5	ICS121	Data Structures I	3	1	2	5
IEC111	Electronic Circuits	3	1	2	5	ICS122	Computer Organization	3	1	0	4
IHS111	Communication Skills	3	0	0	3	ICS123	IT Workshop II	2	1	2	4
IHS112	Foreign Language	1	0	0	1	IHS121	Personality Development	1	0	0	1
IPT111	Physical Training I	0	0	2	0						
Total		16	4	8	23	Total		15	5	6	23
Cumulative Credits at the End of First Year: 46											
Semester -III						Semester -IV					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA211	Probability, Statistics and Random Processes	3	1	0	4	IMA221	Differential Equations and Transforms	3	1	0	4
ICS211	Design and Analysis of Algorithms	3	1	0	4	ICS221	Operating Systems (2024 onwards)	3	1	0	4
ICS212	Theory of Computation (2024 onwards)	3	1	0	4	ICS223	Compiler Design	3	0	2	4
ICS213	Database Management Systems	2	1	2	4	ICS224	Computer Networks	3	0	2	4
ICS214	IT Workshop III	2	1	2	4	ICS225	Data Structures III	1	0	2	2
ICS215	Data Structures II	1	0	2	2	ICS226	Secure Software Engineering	3	0	2	4
ISC212	Quantum Computing and Security	2	0	0	2	IHS221	Fundamentals of Economics	1	0	0	1
IPT211	Physical Training II	0	0	2	0	IHS223	Risk Management	1	0	0	1
Total		16	5	8	24	Total		18	2	8	24
Cumulative Credits at the End of Second Year: 94											
Semester -V						Semester -VI					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CBE311	Database Security	3	0	0	3	CBE321	Machine Learning and Cyber Security	3	0	0	3
CBE312	Internet of Things and its Security	3	0	2	4	CBE322	Digital Forensics	3	0	2	4
IEG312	Distributed System Security	3	0	2	4	CBS321	Cloud Computing and Security	3	0	2	4
CBS311	Fundamentals of Data Science	3	0	2	4	CBE323	Cryptography	3	0	2	4
CBE312	Internet of Things and its Security	3	0	2	4	ISC322	Criminal Psychology and Behavior Intelligence	3	0	0	3
IMA313	Number Theory and Mathematical Theory of Coding	3	0	0	3	IOE322	Information Security Standards, Policies, Strategies & Audits	3	0	0	3
IHS314	Financial Crime, Motivations and Typologies	3	0	0	3	IHP323	Honours Project Phase I (Optional)				
Total		19	0	6	22	Total		18	0	6	21
Cumulative Credits at the End of Third Year: 137											
Semester -VII						Semester -VIII					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CBS411	Mobile Forensics and Security	3	0	2	4	CBS421	Cyber Ethics, Privacy and Legal Issues	3	0	0	3
CBE411	Vulnerability Assessment and Penetration Testing	3	0	2	4	CBS422	Biometric Security / Industrial Training	3	0	2	4
CBE412	Multimedia Security & Forensics	3	0	2	4	IOE422	Lightweight Cryptography	3	0	0	3
IOE411	Blockchain & Crypto-Currencies	3	0	0	3	IBP423	B. Tech Project Phase II	6	0	0	6
IBP413	B. Tech Project Phase I	6	0	0	6	IHP423	Honours Project Phase II (Optional)				
CBSXXX	Research Course (Optional)					CBSXXX	Research Course (Optional)				
Total		18	0	6	21	Total		15	0	2	16
Cumulative Credits at the End of Fourth Year: 174 (B.Tech); 174 + 12 = 186 (B.Tech (Hon)); 174 + 20 = 194 (B.Tech-MS)											
Semester -IX						Semester -X					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IRPXXX	Research Project	12	0	0	12	IRPXXX	Research Project	12	0	0	12
Total		12	0	0	12	Total		12	0	0	12
Cumulative Credits at the End of Fifth Year: 218 (B.Tech-MS)											

Remark: To meet the minimum requirement of 186 credits for qualifying the B.Tech (Hon)Degree, students may take two additional projects of 6 credits each and, to meet the requirement of 218 credits for B.Tech-MS, students may take two additional projects of 6 credits each, two 4 credit research courses and 24 credit research project in addition to 174 credits requirement of B.Tech Degree.

B.TECH./B.TECH (HON)/DUAL DEGREE (B.TECH - MS) PROGRAMME

Sl. No.	Course Description	Minimum Credits Requirement			Period
		B.Tech	B.Tech(Hon)	B.Tech-MS	
1	Institute Core/Foundation Courses	94	94	94	Semester I to IV
2	Bouquet Core Courses	24	24	24	Semester V to VIII
3	Stream Electives	18	18	18	Semester V to VIII
4	Institute Open Electives	9	9	9	Semester V to VIII
5	Humanities Electives	3	3	3	Semester V to VIII
6	Science Electives	3	3	3	Semester V to VIII
7	Maths Elective	3	3	3	Semester V to VIII
8	Engineering Elective	4	4	4	Semester V to VIII
9	Any other Electives/Industrial Training	4	4	4	Semester V to VIII
10	B.Tech Projects	12	12	12	Semester VII to VIII
11	Honours Project	-	12	12	Semester VI to VIII
12	Research Courses	-	-	8	Semester VI to VIII
13	Research Project	-	-	24	Semester IX to X
Total Credits required for Successful Completion		174	186	218	
Minimum CGPA required for Successful Completion		5.5	8.0	8.0	

SEMESTER I

IMA111 Discrete Mathematics [3-1-0-4]

Course Objectives

- To extend student's Logical and Mathematical maturity and ability to deal with abstraction
- To introduce most of the basic terminologies used
- To explain and apply the basic methods of discrete mathematics
- To able to write clear, concise and correct mathematics proofs.
- To solve counting problems involving permutations and combinations and apply Pigeon hole principle
- To understand the basics of graph theory and group theory

Course Outcomes

- Have knowledge of the concepts needed to test the logic of a program.
- Have an understanding in identifying structures on many levels.
- Be aware of a class of functions which transform a finite set into another finite set which relates to input and output functions in computer science.
- Be able to apply basic counting techniques to solve combinatorial problems
- Acquire ability to describe computer programs in a formal mathematical manner.

Syllabus

Logic: Propositions, negation, disjunction and conjunction, implication and equivalence, truth tables, predicates, quantifiers, rules of inference, methods of proof.

Set theory: definition and simple proofs in set theory, Inductive definition of sets and proof by induction, inclusion and exclusion principle, relations, representation of relations by graphs, properties of relations, equivalence relations and partitions, partial orderings.

Functions: mappings, injection and surjections, composition of function, inverse functions, special functions, recursive function theory.

Elementary combinatorics: Counting techniques, pigeon-hole principle, recurrence relation, generating functions.

Graph theory: Elements of graph theory, Euler graph, Hamiltonian path, trees, tree traversals, spanning trees.

Algebra: groups, Lagrange's theorem, homomorphism theorem, rings and fields, structure of the ring $\mathbb{Z}_n$ and the unit group $\mathbb{Z}_n^*$, lattice.

Textbooks/References

1. Kenneth H. Rosen, Discrete Mathematics and Its Applications, Seventh Edition, McGrawHill, 2017.
2. Norman L. Biggs, Discrete Mathematics, Oxford University Press, Second Edition, 2003.
3. J.P. Tremblay, R. Manohar, Discrete Mathematical Structures with applications to Computer Science, McGraw Hill, 2017.
4. K.A. Ross, C.R. B. Wright, Discrete Mathematics, 5th Edition, Pearson, 2003.
5. P. B. Bhattacharya, S. K. Jain, S. R. Nagpaul, Basic Abstract Algebra, Second Edition, Cambridge University Press, 2003.
6. J.A. Gallian, Contemporary Abstract Algebra, Ninth Edition, Cengage Learning, 2017.

ICS111 IT Workshop I [3-1-2-5]

Course Objectives

- To extend student's knowledge in basics of computers and networks.
- To enable students to understand the internals of systems, which includes hardware parts, software and its installation procedure, basic Linux commands.
- Students would also be getting familiar with various networking terminologies and its components.
- They would be also studying to setup and develop a dynamic web application.

Course Outcomes

- Have a knowledge of the various hardware components.
- Understand Linux commands and shell scripting.
- Be aware of basic networking concepts, devices and its functionality.
- Be aware of the basic web development scripting languages like HTML, CSS, XML, and JavaScript.

Syllabus

Computer Hardware – Prerequisite, Computing Agents CPU, Memory, Motherboard - Computer Peripherals - I/O devices, Storage devices, Interface cards – Buses – Firmware - Boot process.

Computer System Software - Operating Systems, Unix/Linux commands, Shell scripting. Computer Communications – LAN, MAN, WAN, Client/Server networks, Peer-to-Peer networks, Topologies, Basics of IP addresses, DNS, Routers, Internet, WWW, Web servers.

Web Design - Basics of HTML, CSS and JavaScript, Dynamic Webpage creation, Angular JS: The basics of AngularJS, Introduction MVC, Data Binding, Filters and modules, Directives, Working with Forms, Validation, Angular JS animation, Deployment considerations.

Lab Practice

Computer Hardware – Familiarization CPU Box, Mother board, CPU & Chip-set, Interface cards, Card slots, Hard disk, Cables, SMPS, NIC, Various ports, etc.

Computer Peripherals - I/O Devices. Storage devices, Interface cards – Buses – Firmware - Boot process - Writing/formatting media.

Computer System Software - Operating Systems, Windows, Linux, Commands Unix/Linux commands, Shell scripting. Client/Server networks, Peer-to-Peer networks, LAN, WAN, MAN.

Familiarization of - Basics of TCP/IP, IP addresses, DNS, Routers, Internet, WWW, FTP, Email servers, Web servers.

Web Design - Basics of HTML, CSS, XML, Java Scripting

Textbooks/References

1. Kevin Wilson, Exploring Computer Systems: The Illustrated Guide to Understanding Computer Systems, Hardware & Networks, Volume 6 of Exploring Tech, Elluminet Press, 2019.
2. Craig Zacker, John Rourke, PC Hardware: The Complete Reference, McGraw Hill Education, 1st edition, 2017.
3. Shotts, W, The Linux command line: a complete introduction. 2nd Ed., No Starch Press, 2019
4. Alan Clements, Principles of Computer Hardware, Oxford University Press India, Fourth Edition, 2013.
5. Robbins, J. N. (2012). Learning web design: A beginner's guide to HTML, CSS, JavaScript, and web graphics. "O'Reilly Media, Inc."
6. Meloni, Julie C., Kyrnin, Jennifer. HTML, CSS, and JavaScript All in One, Sams Teach Yourself. United Kingdom: Pearson Education, 2018.
7. Duckett, Jon. Web design with HTML, CSS, JavaScript and jQuery set. Vol. 1. IN: Wiley, 2014.
8. Andrew Grant, Beginning AngularJS. Apress, 2014.
9. Angular: Up and Running: Learning Angular, Step by Step, O'Reilly; 1st edition, 2018.

ICS112 Computer Programming [3-1-2-5]

Course Objectives

- To introduce the problem-solving processes and techniques
- To teach computer programming
- To use C & C++ for solving the problems

Course Outcomes

1. Students learn how to write the sequence of operations in solving a problem
2. Students learn to translate the problem solving steps to a program
3. Students learn the use of programming language for solving real world problems on a computer

Syllabus

Basics of computers: software/ systems, Programming- Introduction, Problem solving, Introduction, Problem solving techniques: definition of problems, solutions, top-down approach, breaking problem in to sub-problems.

Algorithms: - writing the steps required solving problems, representing algorithms as flowchart, translating to procedure/ functions. Modularity

Example problems: computation of factorial, sine, Mod arithmetic computation of quotient/ remainder, solving factorial through recursion, etc. Object technology- introduction, C++ data types/ scope rules, C++ control statements, Example problems/ program, Example problems/ program, Character handling, Pointers, functions, Classes and objects, Classes and objects.

Lab Practice

1. Implement fundamental domain knowledge of the course for developing effective computing solutions by incorporating creativity and logical reasoning.
2. Students are encouraged to use the lab sessions as a multi-use, technology-enhanced teaching space with characteristics of both classrooms and labs.
3. Understand and learn how a big program can be broken up into independent modules and define functions and call them with appropriate parameters.
4. Students should gain a clear idea of how decision making and various basic/advanced constructs for control flow and instruction repetition is done while programming.
5. Students should learn how to use arrays for storing/retrieving large amount of data. They should also understand the concept of strings and string libraries used for their manipulation.
6. Comprehend how to use structures as a compound datatype. Students should also acquire the capability to design structures according to their requirement.
7. Understand recursion, pointer referencing/dereferencing and dynamic allocation of memory.

Textbooks/References

1. R G Dromey, How to Solve It by Computer, Prentice-Hall International Series in Computer Science, 2006.
2. G. Michael Schneider, Invitation to Computer Science, Eighth Edition, 2018.
3. Byron S Gotfried, Programming with C, Third Edition, McGraw Hill Companies, 2017.
4. Michael Vine, C Programming for the Absolute Beginner, Third Edition, 2014.
5. Brian W Kernighan, Dennis M. Ritchie, C Programming Language, Second Edition, Pearson Education India, 2015.
6. Herbert Schildt, C++ Complete Reference, McGraw Hill, Fourth Edition, 2017.
7. Eric Nagler, Learning C++ : A hands-on Approach, Third Edition, Cengage learning, 2017.

IEC111 Electronic Circuits [3-1-2-5]

Course Objectives

- To impart the basic concepts of semiconductor devices.
- To make students capable of analyzing the operation of various electronic circuits.
- To develop knowledge for designing simple analog electronic circuits using discrete and integrated components.

Course Outcomes

- Ability to analyze PN junctions in semiconductor devices under various conditions.
- Ability to design and analyze simple rectifier/clipper/clamper circuits using diodes.
- Ability to design and analyze amplifier circuits using BJTs and MOSFETs.
- Ability to simulate simple electronic circuits using softwares such as LTSpice/PSpice .

Syllabus

Semiconductors: Intrinsic semiconductors, doped semiconductors, current flow in semiconductors, drift current, diffusion current, the pn junction, physical structure, semiconductor diode, diode equation, diode characteristics, static and dynamic resistances, diode equivalent models: ideal diode model, piecewiselinear model, constant voltage-drop model, diode applications: rectifiers, clipping and clamping circuits, zener diodes as regulators.

Bipolar Junction Transistors (BJTs): Introduction to transistors, transistor construction, operation, NPN and PNP transistors, transistor voltages and currents, transistor characteristics. Common-base and common-emitter transistor configurations, biasing BJTs: fixed-bias, emitter-stabilized bias, voltage-divider bias, dc bias with voltage feedback, emitter follower bias, current

mirror, BJT as amplifiers, voltage gain, current gain, input resistance and output resistance using re model, BJT as a switch.

Field Effect Transistors: Introduction, MOSFET, device structure and physical operation, regions of operation, current-voltage characteristics, body effect, biasing of MOSFET. Optoelectronic devices: photodiodes - current and voltage in an illuminated junction, photodetectors. Light Emitting Diode (LED).

Lab Practice

Familiarization of basic electronic lab equipments and components, diode characteristics, clippers, clampers, rectifiers, zener regulator, BJT input and output characteristics, MOSFET input and output characteristics, simulation of simple diode and transistor circuits.

Textbooks/References

1. Sedra A. and Smith K. C, Microelectronic Circuits", Oxford University Press, Seventh Edition, 2015.
2. Robert. L. Boylestad, Louis Nashelsky, Electronic Devices and Circuits Theory, Pearson Education, Eleventh Edition, 2015.
3. David A Bell, Electronic Devices and Circuits, Oxford University Press, Fifth Edition, 2008.
4. Jacob Millman and Christos. C. Halkias, Electronic Devices and Circuits, Mc. Graw Hill, Fourth Edition, 2015.
5. Ben. G. Streetman, Sanjay Kumar Banerjee, "Solid State Electronic Devices", 7th Edition, Pearson Education, 2016, ISBN 978-93-325-5508- 2.
6. Donald A Neamen, Dhrub Biswas, "Semiconductor Physics and Devices", 4th Edition, MCGraw Hill Education, 2012, ISBN 978-0-07-107010-2.
7. S. M. Sze, Kwok K. Ng, "Physics of Semiconductor Devices", 3rd Edition, Wiley, 2018.

IHS111 Communication Skills [3-0-0-3]

Course Objectives

- The objective of the class was to improve the English communication skills of First Semester B. Tech students who had just passed out of their Senior Secondary classes. This was challenging since the class included students from various parts of the country speaking various mother tongues.
- The syllabus is designed to give importance to essential grammar, as well as reading, writing, and speaking skills.
- Work in class consisted of teaching grammar, interspersed with written exercises, reading practice, reading comprehension, business letter writing, report writing, training in preparing CVs for job applications, group discussion, and extempore speaking.

- The classes will be rounded off with some training in the so-called “soft skills”.

Course Outcomes

- Improvement in English language ability was noted in most of the students.
- A large number showed very good improvement, while even the least competent registered some improvement.
- Under the circumstances, the objective of the class would appear to have been achieved.

Syllabus

Communication: verbal and non-verbal, Conversation: formal and informal, prepared and extempore, English: British/American/Indian, Vocabulary development: reading, use of dictionaries, Expression: writing, Pronunciation: phonetics, use of phonetic dictionaries, speaking.

English grammar: Basics: Parts of speech: Noun, Pronoun, Adjective, Verb, Adverb, Preposition, Conjunction, Interjection. Verb: Tenses. Sentence structure: S+V. Concord: Subject-Verb agreement. Reported speech, Active and passive voice, Tag questions, Confusing words and expressions, Synonyms and antonyms, Idioms and phrases, Common errors in English, Punctuation.

Writing skills: Letters: Formal/Informal, Reports, CV. Comprehension: Listening/Reading/Making notes/Summarizing, Interview skills, Group discussion, Soft-skills.

Textbooks/References

1. A.J. Thomson & A.V. Martinet, *A Practical English Grammar*. Delhi: OUP.
2. George Yule, *Oxford Practice Grammar: Advanced*. Oxford: OUP.
3. Raymond Murphy, *Essential English Grammar*. Delhi: Cambridge University Press.
4. Matthew Monippally, *The Craft of Business Letter Writing*. New Delhi: Tata McGraw-Hill.

IHS112 Foreign Language [1-0-0-1]

Course Objectives

- To introduce the basics of the German language and its grammar to the students.
- To develop the four fundamental language skills at the initial level: listening, speaking, reading, and writing.
- To cover the essentials of the German language, such as the German alphabet and pronunciation, core grammar rules, and simple, everyday vocabulary.
- To provide a basic understanding of the German language and the culture of German-speaking countries (Germany, Austria, Switzerland, etc.).

Course Outcomes

Upon successful completion of this course, students will be able to:

- Introduce themselves and others, and ask and answer basic personal questions (e.g., about where they live, people they know, and things they have).
- Understand and use familiar, everyday expressions and very simple sentences aimed at satisfying concrete needs (e.g., ordering food, asking for directions).
- Interact in a simple way, provided the other person talks slowly and clearly and is prepared to help.
- Read and understand the main points in short, simple texts like public notices, posters, or short personal messages.
- Write a short, simple text, such as a postcard with holiday greetings, or fill in forms with personal details like their name, nationality, and address.
- Develop a basic awareness of German culture and daily life.

Syllabus

Introduction and Basics: The German Alphabet and Pronunciation (Das deutsche Alphabet und Aussprache): Learning the sounds of German letters and letter combinations. Greetings and Introductions (Begrüßungen und Vorstellungen): Formal and informal greetings, introducing oneself and others. Numbers, Time, and Dates (Zahlen, Uhrzeit und Datum): Numbers from 0 to 100, telling time, and mentioning dates.

Basic Grammar and Sentence Structure: Nouns, Gender, and Articles (Nomen, Genus und Artikel): Understanding that German nouns have a gender (masculine, feminine, neuter). Using definite and indefinite articles (*der/die/das, ein/eine/ein*) in the Nominative case. Simple Sentence Structure: Forming basic statement sentences and questions (W-Fragen like *Wer? Was? Wo?* and Ja/Nein questions). Verb Conjugation (Present Tense) (Konjugation im Präsens): Conjugating regular verbs and important irregular verbs like *sein* (to be), *haben* (to have), and *sprechen* (to speak). A1 Connectors, Basic writing.

Building Vocabulary and Conversations: Topics: The Family (*die Familie*), Hobbies (*die Hobbys*), Food and Drinks (*Essen und Trinken*), and Professions (*Berufe*). Grammar: The Accusative case for direct objects. Negation with *nicht* and *kein*. Possessive Articles (*Possessivartikel*): my, your, his, her, etc. (e.g., *mein Buch, deine Tasche*).

Expanding Skills: Talking about the Past: Introduction to the Present Perfect tense (*Perfekt*) to talk about past events (e.g., “I have learned German”). Separable Verbs (*Trennbare Verben*): Understanding verbs like *anrufen* (to call) or *einkaufen* (to shop). Imperative sentence formation.

Textbooks/References

1. *Netzwerk neu A1* (Kursbuch und Arbeitsbuch) by Klett Sprachen.
2. *Netzwerk neu A1* (Vocabulary) by Klett Sprachen.
3. *Grammatik aktiv A1-B1* by Cornelsen.

SEMESTER II

IMA121 Calculus and Linear Algebra [3-1-0-4]

Course Objectives

- To study the basic topological properties of the real numbers.
- Have the knowledge of the sequence of real numbers and convergence.
- Studying the notion of continuous functions and their properties.
- To gain an understanding of the linear system of equations.
- To get introduced to the fundamental concepts of vector spaces.
- To impart the basics of linear transformation, orthogonalization, basis, dimensions, and eigenvalues.
- To provide the knowledge to apply the concepts of linear algebra in engineering applications.

Course Outcomes

- Have a good knowledge of the mathematical concepts in real analysis.
- Be able to prove statements and to formulate precise mathematical arguments.
- To solve the problems related to linear systems and matrices.
- To apply the knowledge of linear transformation, orthogonal projections, ortho-normalization, and least-square solutions in engineering applications.

Syllabus

Calculus: Real Numbers, Properties of Real Numbers, Limit, Continuity, Discontinuity, Uniform continuity, Differentiability, Rolle's theorem, Mean Value theorem, Riemann Integration, Fundamental theorem of calculus.

Sequences and Series: Convergence and limit laws, Finite and infinite series, Sums of non-negative numbers, Absolute and conditional convergence of an infinite series, Tests of convergence.

Function of Several Variables: Limit, Continuity, Partial derivatives, Local maxima and local minima, Saddle point, Hessian Matrix.

Vector Calculus: Parameterized curves in space, Arc length, Tangent and normal vectors, Line integral, Gradient, Directional derivatives, Tangent plane and normal vector, Vector field, Divergence, Curl, Parameterized surface, Surface integral, Integral theorems (statements only): Green's Theorem, Stokes' theorem, Gauss' divergence theorem. Coordinate system: Polar, Spherical, Cylindrical.

Linear Algebra: System of linear equations, Row reduced echelon matrices, Rank of a matrix. Definition of a linear vector space and examples; Linear independence of vectors,

Basis and dimension, Subspaces; Linear transformations, Matrix representation of Linear transformation, Inner product, Norms, Orthogonal basis, Gram-Schmidt orthogonalization process, Eigenvectors of a matrix and matrix diagonalization, Applications.

Textbooks/References

1. R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, Fourth Edition, Wiley, 2011.
2. T. M. Apostol, *Calculus, Volume I*, Second Edition, Wiley, 2007.
3. Gilbert Strang, *Linear Algebra and Its Applications*, Fifth Edition, Wellesley-Cambridge Press/SIAM, 2016.
4. K. Hoffman and R. Kunze, *Linear Algebra*, Second Edition, PHI, 2009.
5. Erwin Kreyszig, *Advanced Engineering Mathematics*, Tenth Edition, Wiley, 2015.

IEC121 Digital Design and Electric Circuits [3-1-2-5]

Course Objectives

- The primary objective of this course is to provide the student with the fundamental concepts and skills necessary to analyze and design combinational and sequential logic circuits.
- The course explains the basics of analog and digital logic circuits.
- It also introduces the student to a hardware description language and its application to the design of combinational, sequential, and simple digital systems.
- The material covered in the lecture is reinforced through practical experience in the associated lab together with the use of Verilog HDL to synthesize logic circuits.

Course Outcomes

- Understand the fundamentals of analog and digital circuits.
- Analyze and design a circuit of logic gates that have the desired relation between the input and output terminals.
- Understand the logic properties of flip-flops.
- Analyze and design counters, registers, and similar circuits.
- Implement combinational and sequential circuits using a hardware description language.

Syllabus

Introduction - Analog and Digital circuits – Kirchhoff's Laws, Superposition Theorem, Thevenin's and Norton's Theorems; Review of Number Systems - Number systems and conversions-decimal, binary, 1's and 2's complements,

hexadecimal, octal etc. Logic gates-NOT, AND, OR, XOR, XNOR, Universal gates, timing diagrams.

Boolean algebra-DeMorgans theorems, SOP and POS forms. Karnaugh Maps-to simplify Boolean expressions, truth table functions. Combinational Logic-Analyze basic combinational logic circuits, design a combinational logic circuits for a given truth table. Functions of Combinational logic-comparators, adders, code converters, multiplexers, de-multiplexers.

Sequential Circuit Design - Flip-Flops and Latches. SR, D, and JK Flip-Flops. Edge- triggered and Master-Slave Flip-Flops, Excitation table. Counters – Design of asynchronous and synchronous counters. Timing diagrams up/down counters. Shift Registers – data movements in shift registers. SISO, SIPO, PISO, PIPO shift registers.

Memory and programmable logic – RAM, Memory decoding, ROM, PLA, PAL, sequential programmable devices, overview of logic design using Verilog HDL, Basic concepts, Modules, Ports.

Lab Practice

- Familiarization of Logic Gates.
- Design of Combinational Logic Circuits – Comparators, Adders, Code Converters, Multiplexers, Demultiplexers etc.
- Familiarization of Flip-Flops and Latches. SR, D, and JK Flip-Flops. Edge- triggered and Master-Slave Flip-Flops.
- Design of Sequential Logic Circuits, Design of Counters, Asynchronous Counters, Synchronous counters. Shift Registers.
- Simple Verilog HDL programs.

Textbooks/References

1. Floyd, Digital Fundamentals, McGraw Hill, Tenth Edition, 2011.
2. Morris Mano, Digital Circuits and Logic Design, PHI Publication, Fifth Edition, 2015.

ICS121 Data Structures I [3-1-2-5]

Course Objectives

- Define and describe simple data structures like arrays, linked lists, trees, and graphs.
- Design and specify algorithms for searching and sorting, and those associated with the above data structures.
- Analyze simple algorithms, like sorting and searching using mathematical tools, like formulation and solving of recurrences, asymptotic analysis, and probabilistic analysis.
- Analyze application problems and abstract them to formulate solutions involving data structures and algorithms.

Course Outcomes

- Students learn to define operations of data structures like arrays, linked lists, trees, and graphs.

- Students learn to design and specify algorithms involving the above types of data structures.
- Students learn to analyze simple algorithms and solve recurrences, asymptotic analysis, and probabilistic analysis.
- Students learn to analyze application problems and abstract them to formulate solutions involving data structures and algorithms.

Syllabus

Introduction- Algorithm Analysis, Finding Complexity. Fundamental data structures - List- Sorted Lists, Double Linked Lists, Skip list.

Stack & Queue application - Celebrity problem, histogram rectangular area problem.

Binary Trees – Insertion and Deletion of nodes, Tree Traversals, Polish Notations, Red Black Trees, B-Trees, Heaps, Priority Queues.

Optimal binary search tree, Application problems on Optimal binary search Tree.

Sorting – Bubble, Selection, Insertion, Merge Sort, Quick Sort, Radix Sort, Heap sort. Searching.

Hashing- Application problems on hashing.

Graphs- Shortest path algorithms, Minimum Spanning Trees, BFS, DFS.

Textbooks/References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goodwasser, Data Structures And Algorithms In Java™ Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss, Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

ICS122 Computer Organization [3-1-0-4]

Course Objectives

- To understand the basics of computer hardware and how software interacts with computer hardware.
- To analyze and evaluate the performance of computers.
- Understand basics of Instruction Set Architecture (ISA) – RISC.
- To understand how computers represent and manipulate data.

- To understand how computers perform arithmetic operations, how they are optimized and made to run faster.
- To understand how memory management takes place in a computer system.
- To understand what pipelining is, and the design concepts involved.
- Design a simple computer with hardware design including data format, instruction format, instruction set, addressing modes, bus structure, input/output, memory, Arithmetic/Logic unit, control unit, and data, instruction, and address flow.

Course Outcomes

- This course will introduce students to the fundamental concepts underlying modern computer organization and architecture.
- Students should be able to understand the overall working of a computer.
- Students should gain a detailed understanding of the design principles involved in developing a computer.
- They should know the representation of data, how programs are represented, executed, and how programs manipulate and operate on data.
- They should also be able to appreciate how memory organization is done and how to organize memory for faster execution of programs.
- Students should also be able to appreciate the concepts in pipelining.

Syllabus

Computer abstraction and technology: Basic principles, hardware components, Measuring performance: evaluating, comparing, and summarizing performance. Instructions: operations and operands of the computer hardware, representing instructions, making decisions, supporting procedures, character manipulation, styles of addressing, starting a program.

Computer Arithmetic: signed and unsigned numbers, addition and subtraction, logical operations, constructing an ALU, multiplication and division, floating point representation and arithmetic, Parallelism and computer arithmetic.

The processor: building a data path, simple and multi-cycle implementations, microprogramming, exceptions, Pipelining, pipeline Data path and Control, Hazards in pipelined processors.

Memory hierarchy: caches, cache performance, virtual memory, common framework for memory hierarchies. Input/output: I/O performance measures, types and characteristics of I/O devices, buses, interfaces in I/O devices, design of an I/O system, parallelism and I/O. Introduction to multicores and multiprocessors.

Textbooks/References

1. D. A. Patterson and J. L. Hennessy, Computer Organisation and Design: The Hardware/ Software Interface, Fourth Edition, Morgan Kaufman, 2009.
2. V. P. Heuring and H. F. Jordan, Computer System Design and Architecture, Prentice Hall, 2003.
3. J. L. Hennessy and D. A. Patterson, Computer Architecture: A Quantitative Approach, Fifth Edition, Morgan Kaufman, 2011.
4. Carl Hamacher, Zvonko Vranesic, and Safwat Zaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

ICS123 IT Workshop II [2-1-2-4]

Course Objectives

- To introduce the object-oriented problem-solving processes/ techniques.
- To learn Object-oriented programming.
- To use JAVA application development platforms.

Course Outcomes

- Students learn OOP development on Eclipse and developer platform.
- Students learn to write larger complex applications.

Syllabus

Introduction to OOPs and Java Language Fundamentals - OOPs Principles, Features of Java. JVM, Bytecode, JRE, Language Fundamentals.

Classes and Objects- Introducing class fundamentals, Object and Object reference, Introducing methods, Extending Objects, Object Lifetime & Garbage Collection, Creating and Operating Objects, Constructor & initialization code block, Access Control, Modifiers, Nested, Inner Class and Anonymous Classes, Abstract Class and Interfaces, Defining Methods, Argument Passing Mechanism, Method Overloading, Recursion, Dealing with Static Members, finalize method, native Method. Use of **this** reference, Use of Modifiers with Classes & Methods, Method overriding, Garbage Collection.

Extending Classes and Inheritance: Use and Benefits of Inheritance in OOP, Types of Inheritance in Java, Role of Constructors in inheritance, Overriding Super Class Methods, use of **super**, Polymorphism in inheritance, Implementing interfaces.

Package Organizing Classes and Interfaces in Packages, Package as Access Protection.

Exception Handling: Exceptions & Errors, Types of Exception, Control Flow in Exceptions.

Array & Strings: Defining an Array, Initializing & Accessing Array, Multi-Dimensional Array, Operation on String,

Mutable & Immutable String, Using Collection Based Loop for String, Tokenizing a String, Creating Strings using String Buffer.

Introduction to PHP app development- Introduction to PHP, Basic Syntax of PHP, PHP statement terminator and case insensitivity, Embedding PHP in HTML, Comments, Variables, Assigning value to a variable, Constants, Managing Variables; Operators and Controls Structures; Functions in PHP, Arrays, PHP File and Forms Handling- PHP Server-side programming - Using PHP With HTML Forms, GET and POST methods, Sessions and Cookies, Support for Database, Creating classes in PHP.

Textbooks/References

1. C. Thomas Wu, An Introduction to Object-Oriented Programming with Java, Fifth Edition, 2009.
2. Ken Arnold, James Gosling and David Holmes, The Java Programming Language, Fourth Edition, 2005.
3. Herbert Schildt, Java: The Complete Reference, McGraw Hill Education (India), 11th Edition 2018.
4. Steven Holzner, PHP: The Complete Reference, McGraw Hill Education, 2017.

The Traits Approach; Models of healthy personality: the notion of the mature person, the self-actualizing personality, etc. Personality disorders; Psychotherapeutic techniques and Yoga & Meditation; Indian perspective on personality; Personality in the socio-cultural context.

Textbooks/References

1. Schultz, D.P., & Schultz, S. E. (2005) (8th Edn.) Theories of Personality. Belmont: Thomson Wadsworth.
2. Lindzey, G., Campbell, J.B., & Hall, C.S. (2007) (4th Edn.). Theories of Personality. New York: Wiley & Sons.
3. Ryckman, R.M. (2008) (9th Edn.). Theories of Personality. Belmont: Thomson Wadsworth.
4. Rao, K.R., & Paranjpe, A.C. (2016). Psychology in the Indian Tradition. New Delhi: Springer.
5. Frankl, V.E. (1992). Man's Search for Meaning. Massachusetts: Beacon Press.
6. Simanowitz, V., & Pearce, P. (2003). Personality Development. England: Open University Press.

IHS121 Personality Development [1-0-0-1]

Course Objectives

- To understand the basic perspectives of human personality such as Trait approach, Psychoanalytic approach, Biological basis, Humanistic/phenomenological approach, Behaviorist/learning, Cognitive approach, Interaction perspective, and Transpersonal perspective (Indian and Yoga Psychology).
- Learn to objectively assess and explain the behavior of other people, identify personality traits to predict how a person will behave, and function effectively.
- Understand how hiring decisions are taken based on personality characteristics that serve as requirements of a job.
- Understanding the application of assessment of Type A & B personality on personal health & achievement.
- Understand Personality disorders and their identification. Cognitive Behaviour Therapy in the context of Psychotherapy for personality disorders.

Course Outcomes

Understanding one's own personality and that of others, appreciating the uniqueness of individuals, adapting to people and situations effectively, assessing self and others using scientific tools of personality, coping with challenges in life with a better understanding of human behavior science.

Syllabus

Personality: Meaning & Assessment. Psychoanalytic & Neo-Psychoanalytic Approach; Behavioural Approach; Cognitive Approach; Social-Cognitive Approach; Humanistic Approach;

SEMESTER III

IMA211 Probability, Statistics and Random Processes [3-1-0-4]

Course Objectives

- To expose the students to the modern theory of probability, concept of random variables and their expectations.
- To introduce various discrete and continuous distributions and the concept of estimation theory, confidence interval.
- To illustrate the concept of hypothesis testing, tests for means and variances, Goodness of fit tests.
- To introduce the concept of random processes, Markov chains, Brownian Motion.

Course Outcomes

- Define and apply the concepts of probability and conditional probability.
- Define and illustrate discrete and continuous random variables, their probability mass functions and probability density functions.
- Understand the concept and need of hypothesis testing.
- Perform the tests for means and variances and Goodness of fit test.
- Understand the concept of random processes, Markov chains, Brownian motions.

Syllabus

Axiomatic construction of the theory of probability, independence, conditional probability, and basic formulae.

Random variables and distributions: Univariate, Bivariate and multivariate random variables, Cumulative and marginal distribution function, Conditional and multivariate distributions, Functions of random variables: Sum, product, ratio, change of variables.

Mathematical expectations, moments, moment generating function, characteristic functions; Discrete/continuous distributions and limit theorems: Binomial distribution, Geometric distribution, Poisson distribution, Normal distribution, Exponential distribution, Gamma distribution, Beta distribution, Central limit theorem, Chebyshev's inequality, Law of large numbers.

Estimation Theory: Bias of estimates, Confidence intervals, Minimum variance unbiased estimation, Bayes' estimators, Moment estimators, Maximum likelihood estimators, Chi-square distribution, Confidence intervals for parameters of normal distribution.

Hypothesis testing: Tests for means and variances, hypothesis testing and confidence intervals, Bayes' decision rules, Power of tests, Goodness-of-fit tests, Kolmogorov-Smirnov Goodness-of-fit test.

Definition and classification of random processes, discrete-time Markov chains, Poisson process, continuous-time Markov chains, stationary processes, Gaussian process, Brownian motion.

Textbooks/References

1. S. Ross, Introduction to Probability and Statistics for Engineers and Scientists, Third Edition, Elsevier, 2004.
2. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2000.
3. S. M. Ross, Introductory Statistics, Second Edition, Academic Press, 2009.
4. J. Medhi, Stochastic Processes, Third Edition, New Age International, 2009.
5. V.K. Rohati and A.K. Saleh, An Introduction to Probability and Statistics, Third Edition, Wiley Student Edition, 2006.
6. G. R. Grimmett and D. R. Stirzaker, Probability and Random Processes, Oxford University Press, 2001.
7. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, Third Edition, Wiley, 1968.
8. S.M. Ross, Stochastic Processes, Second Edition, Wiley, 1996.
9. C. M. Grinstead and J. L. Snell, Introduction to Probability, Second Edition, Universities Press India, 2009.
10. S. Ross, A First Course in Probability, 10th Edition, Pearson Education, Delhi, 2018.

ICS211 Design and Analysis of Algorithms [3-1-0-4]

Course Objectives

- Analyze the asymptotic performance of algorithms.
- Demonstrate a familiarity with major algorithms and data structures.
- Apply important algorithmic design paradigms and methods of analysis.
- Synthesize efficient algorithms in common engineering design situations.

Course Outcomes

- Argue the correctness of algorithms using inductive proofs and invariants.
- Analyze worst-case running times of algorithms using asymptotic analysis.
- Describe the divide-and-conquer paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm. Synthesize divide-and-conquer algorithms. Derive and solve recurrences describing the performance of divide-and-conquer algorithms.

- Describe the dynamic-programming paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm. Synthesize dynamic programming algorithms and analyze them.
- Describe the greedy paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm. Synthesize greedy algorithms and analyze them.
- Explain the major graph algorithms and their analyses. Employ graphs to model engineering problems when appropriate. Synthesize new graph algorithms and algorithms that employ graph computations as key components, and analyze them.
- Explain what an approximation algorithm is, and the benefit of using approximation algorithms.

Syllabus

Introduction: Efficiency – Run Time & Space. Analyzing an Algorithm – Insertion Sort- Proof of Correctness – Complexity – Asymptotic Notations.

Divide and Conquer: Analyzing Recursive Algorithms – Merge Sort, Recurrence Relations – Binary Search. Solving Divide and Conquer Recurrences – Recursion Tree – Substitution Method – Master Theorem – Applications of the Master Theorem.

Greedy Algorithms: Locally Optimal Solutions – Interval Scheduling – Minimum Spanning Trees. Prim's Algorithm – Locally Modifying Solutions to Build Better Solutions – Exchange Arguments. Dijkstra's Algorithm – Kruskal's Algorithm – Knapsack – Huffman Coding.

Dynamic Programming: Reusing work across sub computations – Definition of Dynamic Programming – Optimal Rod Cut Problem - Optimal Matrix Chain Multiplication - Bellman-Ford Algorithm, Floyd-Warshall Algorithm – Longest Common Subsequence – Machine Scheduling Problem. Application problems- Max Flow problems in flow networks.

Amortized Complexity Analysis – Aggregate Method, Accounting Method, Potential Method, Dynamic Tables – Balanced Trees.

Intractable Problems: Polynomial Time – class P – Polynomial Time Verifiable Algorithms – class NP – NP completeness and reducibility – NP Hard Problems – NP completeness proofs – Approximation Algorithms.

Textbooks/References

1. Thomas H Cormen, Charles E Leiserson, Ronald L Rivest, Clifford Stein - Introduction to Algorithms, MIT Press, Third Edition, 2010.
2. Jon Kleinberg, Eva Tardos, Algorithm Design, Pearson Addison, Wesley, 2013.

ICS212 Theory of Computation [3-1-0-4]

Course Objectives

- Course will provide a formal connection between algorithmic problem solving and the theory of languages and automata and develop them into a mathematical (and less magical) view towards algorithmic design and in general computation itself.
- The course should in addition clarify the practical view towards the applications of these ideas in the engineering part of CS.

Course Outcomes

Students who successfully complete this course will be able to:

- Model, compare and analyse different computational models using combinatorial methods.
- Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
- Construct algorithms for different problems and argue formally about correctness on different restricted machine models of computation.
- Identify limitations of some computational models and possible methods of proving them.
- Have an overview of how the theoretical study in this course is applicable to an engineering application like designing the compilers.

Syllabus

Introduction: Notion of formal language - Strings, Alphabet, Language, Operations, Finite State Machine, definitions, finite automaton model, acceptance of strings and languages, deterministic finite automaton, equivalence between NFA and DFA, Conversion of NFA into DFA, minimization of FSM, equivalence between two FSMs, Moore and Mealy machines.

Regular expressions: Regular sets, regular expressions, identity rules, manipulation of regular expressions, equivalence between RE and FA, interconversion, Pumping lemma, Closure properties of regular sets, regular grammars, right linear and left linear grammars equivalence between regular linear grammar and FA, interconversion between RE and RG.

Context-free grammars: Derivation, parse trees, Language generated by a CFG, Eliminating useless symbols, ϵ -productions, and unit productions, Chomsky Normal Form. Pushdown automata: Definition, instantaneous description as a snapshot of PDA computation, notion of acceptance for PDAs.

Turing machine: Turing machine, definition, model, design of TM, Computable Functions, recursive enumerable language, Church's Hypothesis, Counter machine, types of TMs, RAM machine.

Un-decidability and classes of problems: Chomsky hierarchy of languages, linear bounded automata and context-sensitive language, Grammar, decidability of problems, Universal Turing Machine, un-decidability of Post's correspondence problem, Turing reducibility logical theories.

Complexity classes: P, NP, co-NP, EXP, PSPACE, L, NL,

Textbooks/References

1. M. Sipser, "Introduction to the Theory of Computation," Second Edition, Thomson, 2005.
2. Lewis H.P. and Papadimitriou C.H., "Elements of Theory of Computation," Prentice Hall of India, Fourth Edition, 2007.
3. S. Arora and B. Barak, "Computational Complexity: A Modern Approach," Cambridge University Press, 2009.
4. C. H. Papadimitriou, "Computational Complexity," Addison-Wesley Publishing Company, 1994.
5. D. C. Kozen, "Theory of Computation," Springer, 2006.
6. D. S. Garey and G. Johnson, "Computers and Intractability: A Guide to the Theory of NP-Completeness," Freeman, New York, 1979.
7. J. Hopcroft, J.D. Ullman, and R. Motwani, "Introduction to Automata Theory, Languages and Computation," Third Edition, Pearson, 2008.

ICS213 Database Management Systems [2-1-2-4]

Course Objectives

- To understand the need for a database and its management using DBMS.
- To model Entity-Relationship (ER) diagram for a real-world scenario.
- To write relational algebra and relational calculus queries for data handling and retrieval, Write SQL queries for database creation and analysis.
- Design efficient database systems using the principle of normalization.
- Understand the basics of database transactions, deadlock handling, and security.
- How to implement database indexing.
- Usage of tools like RA Interpreter and MySQL for executing various queries.

Course Outcomes

- Understand database concepts and structures and query language.
- Understand the ER model and relational model.
- Apply various Normalization techniques.
- Understand query processing and techniques involved in query optimization.

Syllabus

Database Modeling: Database System concepts and architecture, Data modeling using Entity Relationship (ER) model and Enhanced ER model, Specialization, Generalization.

Database Indexing: Data Storage and indexing- Single level and multi-level indexing, Dynamic Multi-level indexing using B Trees and B+ Trees.

Relational Databases: The Relational Model, Relational database design using ER to relational mapping, Relational algebra, Relational calculus, Tuple Relational Calculus, Domain Relational Calculus, SQL.

Database Design: Database design theory and methodology, Functional dependencies and normalization of relations, Normal Forms, Properties of relational decomposition, Algorithms for relational database schema design.

Database Transactions: Transaction processing concepts, Schedules and serializability, Concurrency control, Two Phase Locking Techniques, Optimistic Concurrency Control, Database recovery concepts and techniques.

Database Security: Introduction to database security.

Textbooks/References

1. Ramez Elmasri and Shamkant B. Navathe, Fundamentals of Database Systems, Fifth Edition, Pearson Education, 2008.
2. Raghu Ramakrishnan and Johannes Gehrke, Database Management Systems, Third Edition, McGraw Hill, 2014.
3. Peter Rob and Carlos Coronel, Database System- Design, Implementation and Management, Seventh Edition, Cengage Learning, 2007.

ICS214 IT Workshop III [2-1-2-4]

Course Objectives

- Learn Python scripting and the scripting shell.
- Master the basics of programming constructs, like conditions, loops, functions, etc.
- Introduce sequence types in Python like Lists, Tuples, Sets and Dictionaries.
- Be exposed to advanced applications such as databases, networks, etc.

Course Outcomes

- Write python programs for various applications.
- Write Database programs to create, access, modify and update data.

- Write network programs for sending emails, ftp, sockets etc.

Syllabus

Introduction to data types, variables, constants, operators, input-output, basic formatting, running python programs, date and time functions. Conditionals, if statement and variants, relational operators, logical operators.

Iteration and while loops, for loops and range command, random numbers.

File processing, reading and writing files, parsing files, text files and CSV files.

Lists and list processing, list operations, list traversals, tables as lists. Tuples, Maps, Sets and Dictionaries, creation and traversals.

Strings and string processing, string functions, conversions.

User defined Functions, lambda functions, recursive functions, built-in functions, yield statement, parameter passing.

Classes and object-oriented programming, inheritance, associations.

Database processing, creating tables, querying, MySQL and PySqlite.

Network programming, sockets, email sending, ftp Threads and multithreading.

Numpy and applications in matrices, random numbers Scipy, Matplotlib, graphing and charting data.

Trie, stack, Networkx, pattern matching, regular expression.

Introduction to Android app development.

Textbooks/ References

1. Ljubomir Perkovic, Introduction to Computing with Python, Wiley, Second Edition, 2015.
2. Narasimha Karumanchi, Data Structures and Algorithms With Python, Careermonk Publications, 2015.

ISC212 Quantum Computing and Security [2-0-0-2]

Course Prerequisites

Student should have a passing grade in Discrete Mathematics (IMA 111) and Calculus and Linear Algebra (IMA 121) or the instructor's approval.

Course Objectives

- To understand the principle of Quantum computing.
- To design Quantum circuits and get hands-on experience using Qiskit.
- To develop an efficient and secure Quantum cryptosystem.
- To understand Quantum Cryptography and Quantum error correction.

Course Outcomes

Students who successfully complete this course will be able to:

- Develop a basic understanding of Quantum Information and Computation.
- Design Quantum circuits in IBM quantum computers.
- Understand the implication of Quantum Algorithms on classical Cryptosystems.
- Understand Quantum Cryptography and Quantum error correction.

Syllabus

Introduction to Quantum Computing and Information: States, Operators, Measurements-Quantum search, Quantum Entanglement- Quantum Teleportation, Super-dense coding- Quantum gates and circuits.

Quantum Algorithms: Introduction, Deutsch-Jozsa algorithm, Grover's Algorithm, Shor's Algorithm, and their cryptanalytic implications. Classical cryptography, Quantum cryptography.

Quantum error correction - The physics of error generation- Diagnosing error syndromes- Qubit error-correcting code.

Textbooks/References

1. Michael A. Nielsen and Isaac L. Chuang, "Quantum Computation and Quantum Information," Cambridge University Press, 2002.
2. A. O. Pittenger, "An Introduction to Quantum Computing Algorithms," Birkhauser, 2000.
3. Phillip Kaye, Raymond Laflamme, and Michele Mosca, "An Introduction to Quantum Computing," Oxford University Press, New York, 2007.
4. P. W. Shor, "Algorithms for Quantum Computation: Discrete Logarithms and Factoring," Foundations of Computer Science (FOCS) 1994, pages 124-134, IEEE Computer Society Press.
5. L. Grover, "A fast quantum mechanical algorithm for database search," Proceedings of 28th Annual Symposium on the Theory of Computing (STOC), May 1996, pages 212-219.
6. Preskill Lecture notes, available online: <http://www.theory.caltech.edu/~preskill/ph229/>.
7. N. David Mermin, "Quantum Computer Science," Cambridge University Press, 2007.
8. B. S. Schneier, "Applied Cryptography: Protocols, Algorithms, and Source Code in C," 2nd Edition, John Wiley and Sons, New York, 1995.
9. A. Peres, "Quantum Theory: Concepts and Methods," Kluwer, 1993.
10. J. J. Sakurai, "Modern Quantum Mechanics," 2nd Edition, Addison-Wesley, 2011.

11. G. Benenti, G. Casati, G. Strini, "Principles of Quantum Computation and Information," Vol. 1: Basic Concepts, Vol II: Basic Tools and Special Topics, World Scientific, 2004.
12. H.-K. Lo, T. Spiller, S. Popescu, "Introduction to Quantum Computation and Information," World Scientific, 1998.
13. N. D. Mermin, "Quantum Computer Science," Cambridge, 2007.
6. Skiena, Steven S. "The algorithm design manual." Springer International Publishing, 2020.
7. Knuth, Donald Ervin. "The art of computer programming." Vol. 3. Pearson Education, 1997.
8. Nomura, Seiichi. "C Programming and Numerical Analysis: An Introduction." Synthesis Lectures on Mechanical Engineering 2.2 (2018): 1-198.
9. Dasgupta, Sanjoy, C. H. Papadimitriou, and U. V. Vazirani. "Algorithms." 2006.
10. Trefethen, Lloyd N., and David Bau III. "Numerical linear algebra." Vol. 50. Siam, 1997.

ICS215 Data Structures II [1-0-2-2]

Course Objectives

- Teach programming with emphasis on problem solving and introduce Data structures.
- Provide the Foundations of the practical implementation and usage of Algorithms and Data Structures.

Course Outcomes

Students who successfully complete this course will be able to:

- Design correct programs to solve problems.
- Choose efficient data structures and apply them to solve problems.

Syllabus

Problems on Geometric algorithms but not limited to: Klee's Algorithm, Manhattan distance problems, Collinear checking, Identifying Integral points inside a Triangle, Circum-center of a Triangle, Triangular Matchstick Number, area of Circumcircle of an Equilateral Triangle, Number of rectangles in $N \times M$ grid, Area of two overlapping rectangles, Number of unique rectangles formed using N unit squares, Circle and Lattice Points, Pizza cut problem.

Algorithms on Bit Manipulation but not limited to: Letter manipulation problems, k -th bit Manipulation problems, Kernighan's Algorithm to count set bits in an integer.

Discussion on Numerical algorithms but not limited to: Gauss-Jordan Elimination, Matrix Manipulation problems.

Textbooks/References

1. Cormen, Thomas H., et al. "Introduction to algorithms." MIT press, 2009.
2. Aho, Albred V., J. E. Hopcroft, and J. D. Ullman. "Data structures and algorithms." 1983.
3. Drozdek, Adam. "Data Structures and algorithms in C++." Cengage Learning, 2012.
4. Allen, Weiss Mark. "Data structures and algorithm analysis in C++." Pearson Education India, 2007.
5. Kleinberg, Jon, and Eva Tardos. "Algorithm design." Pearson Education India, 2006.

SEMESTER IV

IMA221 Differential Equations and Transforms [3-1-0-4]

Course Objectives

- Find the Fourier series representation of a function of one variable.
- Introduce the Fourier series and its application to the solution of partial differential equations.
- Introduce the concepts of Laplace and Fourier transforms.
- Identify the type of a given differential equation and select and apply the appropriate analytical technique for finding the solution of first-order and selected higher-order ordinary differential equations.
- Introduce students to partial differential equations.
- Introduce students to how to solve linear Partial Differential Equations with different methods.

Course Outcomes

Students who successfully complete this course will be able to:

- Analyse and solve engineering problems using Fourier series.
- Find the Laplace and Fourier transforms of functions of one variable.
- Solve first-order differential equations utilizing the standard techniques for separable, exact, linear, homogeneous, or Bernoulli cases. Find particular solutions when given initial or boundary conditions.
- Find solutions of higher-order linear differential equations.
- Classify PDEs, apply analytical methods, and physically interpret the solutions.

Syllabus

Fourier Series: Dirichlet's conditions – General Fourier series – Odd and even functions – Half range Sine and Cosine series – Complex form of Fourier series – Parseval's identity – Harmonic Analysis. Convergence of Fourier Series, differentiation, and integration of Fourier series.

Laplace and Fourier Transforms: Laplace transform pair, Fourier Integral Theorem – Fourier transform pair - Sine and cosine transforms – Properties – Transform of elementary functions – Convolution theorem – Parseval's identity.

Ordinary Differential Equations: Method of variation of parameters – Laplace transform method – Homogeneous equation of Euler's and Legendre's type – System of simultaneous linear differential equations with constant coefficients.

Partial Differential Equations: Formation – Solutions of first-order equations – Standard types and Equations reducible to standard types – Singular solutions – Lagrange's linear equation – Integral surface passing through a given curve – Classification of partial differential equations - Solution of linear equations of higher order with constant coefficients – Linear

non-homogeneous equations.

Fourier Series Solutions of Partial Differential Equations: Method of separation of variables – Solutions of one-dimensional wave equation and one-dimensional heat equation – Steady-state solution of two-dimensional heat equation – Fourier series solutions in Cartesian coordinates.

Textbooks/References

1. C. Edwards and D. Penney, *Elementary Differential Equations with Boundary Value Problems*, 6th edition, Pearson, 2003.
2. W.E. Boyce and R.C. DiPrima, *Elementary Differential Equations*, 7th Edition, John Wiley & Sons, 2002.
3. Erwin Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2015.
4. Tyn Myint-U, L. Debnath, *Linear Partial Differential Equations for Scientists and Engineers*, 4th Edition, Birkhauser, 2007.

ICS212 Operating Systems [3-1-0-4]

Course Objectives

- To introduce the Fundamental concept of OS, and how OS works.
- To study the Building blocks of OS, Components of OS.

Course Outcomes

- Students will be able to design an OS.
- Students will be able to implement various components of OS.
- Students will be able to implement a small OS.

Syllabus

Operating system overview: Computer System Organization, Operating System structure, operations of OS, process management, memory management, storage management, protection and security, distributed systems.

Processes: Process concept, Process scheduling, Operations on processes, Cooperating processes, inter-process communication.

Threads: Overview, Multi-threading models, threading issues, P threads, Windows XP threads.

CPU Scheduling: Basic concepts, scheduling criteria, scheduling algorithms, multiple-processor scheduling.

Process synchronization: The critical section problem, Peterson's solution, synchronization hardware, Semaphores, Monitors, Synchronization examples.

Deadlocks: Methods for handling deadlocks, Deadlock

prevention, deadlock avoidance, Deadlock recovery.

Memory management: Swapping, Paging, Segmentation, Virtual memory, Demand paging, Page replacement.

I/O Systems: I/O hardware, Application I/O interface, Kernel I/O subsystem, transforming I/O requests to hardware operations.

Textbooks/References

1. William Stallings, Operating Systems: Internals & Design Principles, Pearson, Seventh edition, 2014.
2. Andrew S. Tanenbaum, Modern Operating Systems, Pearson, Fourth Edition, 2016.
3. Charles Crowley, Operating Systems - Design Oriented Approach, McGraw Hill Education, First edition, 2017.
4. Abraham Silberschatz, Galvin, Gagne, Operating System Concepts, Wiley, Ninth Edition, 2016.

ICS223 Compiler Design [3-0-2-4]

Course Objectives

- To introduce the major concept areas in compiler design and know the various phases of the compiler
- To understand the various parsing algorithms and comparison of the same
- To provide practical programming skills necessary for designing a compiler
- To gain knowledge about the various code generation principles
- To understand the necessity for code optimization

Course Outcomes

- Apply the knowledge of LEX, YACC, ANTLR tools to develop a scanner and parser
- Suggest the necessity for appropriate code optimization techniques
- Conclude the appropriate code generator algorithm for a given source language
- Design a compiler for any programming language

Syllabus

Introduction to programming language translation. Lexical analysis: Specification and recognition of tokens.

Syntax analysis: Top-down parsing - Recursive descent and Predictive Parsers. Bottom-up Parsing - LR(0), SLR, and LR(1) Parsers.

Semantic analysis: Type expression, type systems, symbol tables, and type checking. Intermediate code generation:

Intermediate languages. Intermediate representation - Three address code and quadruples. Syntax-directed translation of declarations, assignments statements, conditional constructs, and looping constructs.

Runtime Environments: Storage organization, activation records. Introduction to machine code generation and code optimizations.

Lab Practice

- Generation of lexical analyzer using tools such as LEX.
- Generation of parser using tools such as YACC.
- Creation of Abstract Syntax Tree.
- Creation of Symbol tables and Semantic Analysis.
- Generation of target code.

Textbooks/References

1. Aho A.V., Lam M. S., Sethi R., and Ullman J. D., *Compilers: Principles, Techniques and Tools*, Pearson Education, 2007.
2. Appel A.W, and Palsberg J., *Modern Compiler Implementation in Java*, Cambridge University Press, 2002.
3. W. Appel, *Modern Compiler Implementation in C*, Cambridge University Press, 1998.
4. V. Aho, M. S. Lam, R. Sethi, J. D. Ullman, *Compilers - Principles, Techniques & Tools*, Second Edition, Pearson Education, 2007.

ICS224 Computer Networks [3-0-2-4]

Course Objectives

- The students should understand the layers of networking devices.
- They should be familiar with a few networking protocols.
- They should study the different types of networks and topologies of networks.

Course Outcomes

Students who successfully complete this course will be able to:

- Distinguish the importance of different networking components.
- Understand the functionalities of each networking layer and standards.
- Write simple networking-based programs at real and simulator levels.

Syllabus

Evolution of computer networks: Network Architecture - OSI, TCP/IP models.

Physical and Data link layer: Encoding, Framing, Error detection, HDLC, PPP, sliding window protocols, medium access control, Token Ring, Wireless LAN, Packet Switching.

Network Layer: Internet addressing, IP, ARP, ICMP, CIDR, Routing algorithms (RIP, OSPF, BGP).

Transport Layer: UDP, TCP, flow control, congestion control, Introduction to quality of service.

Application Layer: DNS, Web, HTTP, email, authentication, encryption.

Lab Practice

Unix network measurement and analysis tools, NS3 Socket interface and programming, RPC, RMI, Assignments using Network Simulators.

Textbooks/References

1. L. L. Peterson and B. S. Davie, "Computer Networks: A Systems Approach," Fifth Edition, Elsevier, 2011.
2. A. S. Tanenbaum and D. J. Wetherall, "Computer Networks," Fifth Edition, Pearson, 2011.
3. W. R. Stevens, "UNIX Network Programming, Volume 1: Networking APIs: Sockets and XTI," Second Edition, Prentice Hall, 1998.
4. S. S. Panwar, S. Mao, J. Ryoo, and Y. Li, "TCP/IP Essentials: A Lab-based Approach," Cambridge Press, 2004.
5. J. F. Kurose and K. W. Ross, "Computer Networking: A Top Down Approach," Seventh Edition, Pearson India, 2017.
6. D. E. Comer, "Internetworking with TCP/IP Vol. 1," Sixth Edition, Prentice Hall of India, 2006.
7. B. Forouzan, "Data Communications and Networking," Fifth Edition, Tata McGraw Hill, 2012.
8. "Introduction to Network Simulator NS2," Second Edition, 2011.

ICS225 Data Structures III [1-0-2-2]

Course Objectives

- Ensure that the student evolves into a competent programmer capable of designing and analyzing implementations of algorithms and data structures for different kinds of problems.
- Expose the student to the algorithm analysis techniques.

Course Outcomes

Students who successfully complete this course will be able to:

- Analyze the efficiency of programs based on Complexity.

- Understand the necessary mathematical abstraction to solve problems.

Syllabus

Algorithms on Graph connectivity but not limited to: Tarjan's and Kosaraju's strongly connected components algorithms, Detect cycles in an undirected graph, Degree of vertices in a Graph, Path identification between vertices in Undirected graph. Discussions on Randomized algorithms but not limited to: Reservoir Sampling, Birthday Paradox, Load Balancing problem, Karger's algorithm for Minimum Cut, Freivald's Algorithm to check the product of a matrix, Monte Carlo estimation.

Branch and Bound Algorithms: Knapsack problem, Travelling salesman problem.

Threaded Binary Tree, Splay trees, Foldable binary trees, Additional problems on BST, Binomial heap, Fibonacci heap, Topological sorting, self-organizing tree, segment tree, Binary indexed tree, suffix array and suffix tree, pattern searching, Tribonacci word.

Textbooks/References

1. Cormen, Thomas H., et al. "Introduction to Algorithms." MIT Press, 2009.
2. Aho, Alfred V., J. E. Hopcroft, and J. D. Ullman. "Data Structures and Algorithms," 1983.
3. Drozdek, Adam. "Data Structures and Algorithms in C++." Cengage Learning, 2012.
4. Allen, Weiss Mark. "Data Structures and Algorithm Analysis in C++." Pearson Education India, 2007.
5. Kleinberg, Jon, and Eva Tardos. "Algorithm Design." Pearson Education India, 2006.
6. Skiena, Steven S. "The Algorithm Design Manual." Springer International Publishing, 2020.
7. Knuth, Donald Ervin. "The Art of Computer Programming, Vol. 3." Pearson Education, 1997.
8. Nomura, Seiichi. "C Programming and Numerical Analysis: An Introduction." Synthesis Lectures on Mechanical Engineering, 2018.
9. Dasgupta, Sanjoy, C. H. Papadimitriou, and U. V. Vazirani. "Algorithms," 2006.
10. Trefethen, Lloyd N., and David Bau III. "Numerical Linear Algebra." Vol. 50, SIAM, 1997.

ICS226 Secure Software Engineering [3-0-2-4]

Course Objectives

- Foundation on software engineering.
- Design and implementation of secure software.
- Introduce the role of security in the development lifecycle.

- Learn methodological approaches to improving software security during different phases of the software development lifecycle.
- Understand best security programming practices.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand the basics of software engineering.
- Explain terms used in secured software development and lifecycle processes.
- Incorporate requirements into the secured software development process and test software for security vulnerabilities.
- Identify vulnerable code in implemented software and describe attack consequences.
- Apply mitigation and implementation practices to construct attack-resistant software.
- Apply secure design principles for developing attack-resistant software.

Syllabus

Introduction to software engineering: Scope and necessity; Software life cycle models: Waterfall model, Iterative waterfall model, Prototyping model, Evolutionary model, Spiral model, Agile development methodologies.

Requirement analysis and specification; System Design; Basic concepts in user interface design; Software testing; Software Project Management.

Security as a software issue: Introduction, the problem, Software Assurance and Software Security, Threats to software security, Sources of software insecurity, Benefits of detecting software security vulnerabilities.

Requirements engineering for secure software: Introduction, the SQUARE process model, Requirements elicitation and prioritization.

Secure software architecture and design: Introduction, software security practices for architecture and design.

Security and complexity, Governance, and managing for more secure software.

Textbooks/References

1. Roger S. Pressman, "Software Engineering: A Practitioner's Approach," 7th Edition, McGraw-Hill Higher Education.
2. Ian Sommerville, "Software Engineering," 9th Edition, Pearson Education.
3. Julia H. Allen, "Software Security Engineering," Pearson Education.
4. Jason Grembi, "Developing Secure Software," Cengage Learning.
5. Richard Sinn, "Software Security," Cengage Learning.

IHS221 Fundamentals of Economics [1-0-0-1]

Course Objectives

- To familiarize the participants with concepts and techniques in Economics.
- To make the participants appreciate the applications of core concepts in economics for managerial decision-making.
- To sensitize the participants on how the economic environment affects organizations.

Course Outcomes

Students who successfully complete this course will be able to:

- Analyze the demand and supply conditions and assess the positions of a company.
- Design competition strategies, including costing, pricing, product differentiation, and market environment according to the nature of products and structures of the market.

Syllabus

Introduction to Fundamentals of Economics: Micro & Macro Economics, Managerial Economics – Definition, Nature & Scope, Fundamental concepts in Managerial Economics for decision-making: Incremental Principle, Opportunity Cost, Discounting Principle, Time Concept, Equi-Marginal Principle – Illustrations, Decision Making – Process and Conditions, Difference between Risk & Uncertainty.

Demand Analysis and Forecasting: Meaning of Demand – Types of Demand – Law of Demand & its Exceptions, Elasticity of Demand – Price Elasticity, Income Elasticity, Cross Elasticity, Promotion Elasticity, Applications of the concepts of Elasticity, Demand Forecasting – Process – Statistical & Non-Statistical Techniques, Utility Analysis & Consumer Behaviour – Equilibrium of the consumer using Cardinal & Ordinal Utility (Indifference Curve) Theories.

Supply & Production: Theory of Production – Meaning of Production Function, Production Function with One Variable Input – Law of Variable Proportions – Returns to Scale, Production Function with Two Variable Inputs – Iso-quants – Producers' Equilibrium, Economies of Scale – Types – Economies of Scope, Theory of Costs – Classification of Costs – Short Run & Long Run Cost Curves, Revenue Curves.

Market Structure: Market – Meaning & Elements, Classification of Markets – Markets Based on Competition, Theory of Firm – Profit Maximization Rules, Price & Output Determination under Perfect Competition, Price & Output Determination under Monopoly – Monopoly Price Discrimination, Price & Output Determination under Monopolistic Competition, Price & Output Determination under Oligopoly – Kinked Demand Curve Model Only.

Macro Economic Concepts: National Income Concepts – Measurement of National Income, An Overview of the Financial System in India, An Overview of Fiscal & Monetary Policies in India, Balance of Payments: Causes of Disequilibrium & Remedies, Inflation in India – Causes & Remedies. Free Market Economy & Need for Government Intervention – An Appraisal of Economic Reforms in India.

Textbooks/References

1. Dwivedi D.N, "Managerial Economics," Vikas Publications (ISBN 8125910042).
2. P.L. Mehta, "Managerial Economics: Analysis, Problems and Cases," Sultan Chand & Sons (ISBN 81-7014-386-1).
3. K.K. Dewett, "Modern Economic Theory: Micro & Macro Analysis," Orient Book Distributors, New Delhi.
4. V.L. Mote, "Managerial Economics," Tata McGraw Hill, New Delhi.
5. Gaurav Dutt & Aswani Mahajan, "Dutt & Sundaram's Indian Economy," Sultan Chand & Sons.
1. Darril Gibson, Andy Igonor, "Managing Risk in Information Systems," 3rd Edition, Jones & Bartlett Learning, 2021. ISBN: 9781284183726.
2. Michael E. Whitman, Herbert J. Mattord, "Principles of Information Security," 6th Edition, Cengage Learning, 2018.
3. Douglas Landoll, "The Security Risk Assessment Handbook: A Complete Guide for Performing Security Risk Assessments," 3rd Edition, CRC Press, 2021. ISBN: 9781032041650.
4. Darktrace, "Technology," <https://www.darktrace.com/en/technology/#machine-learning>, accessed November 2018.
5. Van Kessel, P., "Is Cyber Security About More Than Protection?" EY Global Information Security Survey 2018-2019.
6. Johnston, A.C. and Warkentin, M., "Fear Appeals and Information Security Behaviors: An Empirical Study," MIS Quarterly, 2010.
7. Arce I. et al., "Avoiding the Top 10 Software Security Design Flaws," IEEE Computer Society Center for Secure Design (CSD), 2014.
8. Smith, H. J., Dinev, T., & Xu, H., "Information Privacy Research: An Interdisciplinary Review," MIS Quarterly, 2011.

IHS223 Risk Management [1-0-0-1]

Course Objectives

- To introduce the fundamental concepts of cybersecurity.
- To understand best practices and methodologies to identify cyber risk, its assessments, and mitigation plans.
- To provide the industry perspectives of cybersecurity.
- To develop a wholesome understanding of cybersecurity and privacy risks to businesses covering governance, compliance, and risk mitigation, and to closely study certain business domains.

Course Outcomes

Students who successfully complete this course will be able to:

- Demonstrate a solid understanding of the fundamental concepts of cybersecurity, including key terminology, principles, and core technologies.
- Apply best practices and methodologies to identify cyber risks, assess their impact, and develop effective mitigation plans for a variety of cybersecurity threats and vulnerabilities.
- Analyze and interpret industry perspectives on cybersecurity, including current trends, emerging threats, and the evolving landscape of cyber threats.
- Recognize the interconnectedness of cybersecurity and privacy within the broader context of business operations.

Syllabus

Introduction to Cybersecurity: Confidentiality, integrity, and availability. CIA triangle, data breaches.

Risk Management: Cyber Risk Identification, Risk Assessment – Identifying Assets and Activities to be Protected, Identifying and Analyzing Threats, Vulnerabilities, and Exploits.

Risk Control and Risk Mitigation Plans.

Cybersecurity: Industry Perspective – Defense Technologies, Attack, Exploits. Cybersecurity Technologies – Access Control, Encryption.

Foundations of Privacy: Information Privacy, Measurement, Theories. Privacy Regulation – Privacy, Anonymity, Regulation, Data Breach, Data Protection.

Textbooks/References

SEMESTER V

CBE311 Database Security [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Database Management System (ICS 213) or the instructor's approval.

Course Objectives

- Introduce the database and its security issues.
- Compare in detail the various state-of-the-art database security methods and techniques.
- Learn in detail the security features in databases.
- Understand the database monitoring tools.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand and characterize modern techniques of database information security threats and techniques for database security assessment.
- Analyze information in a database to identify information security incidents.
- Understand and use the main tools for database management systems monitoring.
- Apply built-in database functions to enable database integrity support.
- Create a plan for vulnerabilities detection and identification in databases.

Syllabus

Fundamentals of Database Systems: Introduction to the Relational Model - Querying relational data, Form of Basic SQL Query - Examples of Basic SQL Queries.

Introduction to Database Security Issues: Confidentiality, Integrity, Availability (CIA triad) for databases, Security challenges in databases, Common database vulnerabilities - Insider attacks, Active directory attacks, buffer overflow attacks, Multi-layered security architecture for databases

Database Security Methods and Techniques: Authentication and authorization mechanisms, Access control models (Discretionary Access Control, Mandatory Access Control, Role Based Access Control), Data masking and redaction (Static and dynamic techniques), Encryption techniques in databases, Secure database backup and recovery mechanisms.

SQL injection attack: Introduction to SQL injection attacks, Types of SQL injection attack, SQL injection mitigation using secure coding practices, SQL injection attack detection and prevention using database firewall.

Database Auditing and Monitoring: Database monitoring tools, Database auditing and logging, database vulnerability scanners. Penetration testing using sqlmap and burp suite.

Textbooks/References

1. Basta A., Zgola M., "Database Security," 3rd Edition, Cengage Learning, US, 2011.
2. Ron Ben Natan, "Implementing Database Security and Auditing," Digital Press, 2005.
3. Bhavani Thuraisingham, "Database and Applications Security," Auerbach Publications, 2005.
4. Rose Anderson, "Security Engineering: A Guide to Building Dependable Distributed Systems," John Wiley & Sons, 2001.
5. Michael Gertz, Sushil Jajodia, "Handbook of Database Security Applications and Trends," Springer, 2008.
6. Silvana Castano, "Database Security," ACM Press.
7. Alfred Basta, Melissa Zgola, "Database Security," Cengage Learning.

IEC312 Distributed System Security [3-0-2-4]

Course Objectives

- To understand different levels of security threats in distributed systems.
- To learn the security solutions for each layer security threats.
- To design a secure distributed system.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand the threats against distributed systems and how to protect against them.
- Have a foundation for designing and developing secure distributed systems, and for evaluating the security of existing solutions.
- Have knowledge of standards, security protocols, technologies, principles, methods, and cryptographic mechanisms applicable for securing modern distributed systems.
- Have knowledge of common mistakes leading to insecurities in distributed systems.

Syllabus

Introduction: Background, Distributed Systems, Distributed Systems Security, Common Security Issues and Technologies.

Host-Level Threats and Vulnerabilities: Background, Malware, Eavesdropping, Job Faults, Resource Starvation, Privilege Escalation, Injection Attacks.

Infrastructure-Level Threats and Vulnerabilities: Introduction, Network-Level Threats and Vulnerabilities, Grid Computing Threats and Vulnerabilities, Storage Threats and Vulnerabilities, Overview of Infrastructure Threats and

Vulnerabilities.

Application-Level Threats and Vulnerabilities: Introduction, Application-Layer Vulnerabilities.

Service-Level Threats and Vulnerabilities: SOA and Role of Standards, Service-Level Security Requirements, Service-Level Threats and Vulnerabilities, Service-Level Attacks, Services Threat Profile.

Host-Level Solutions: Sandboxing, Virtualization, Resource Management, Proof-Carrying Code, Memory Firewall, Antimalware.

Infrastructure-Level Solutions: Network-Level Solutions, Grid-Level Solutions, Storage-Level Solutions.

Application-Level Solutions: Application-Level Security Solutions.

Service-Level Solutions: Services Security Policy, SOA Security Standards Stack, Deployment Architectures for SOA Security.

Textbooks/References

1. Belapurkar, A., Chakrabarti, A., Ponnappalli, H., Varadarajan, N., Padmanabhuni, S., & Sundarrajan, S. (2009). *Distributed Systems Security: Issues, Processes and Solutions*. John Wiley & Sons.
2. Suri, N. (2019). *Distributed Systems Security Knowledge Area*.
3. Firdhous, M. (2012). *Implementation of Security in Distributed Systems-A Comparative Study*. arXiv preprint arXiv:1211.2032.
4. Burns, B. (2018). *Designing Distributed Systems: Patterns and Paradigms for Scalable, Reliable Services*. O'Reilly Media, Inc.
5. Tanenbaum, A. S., & Van Steen, M. (2007). *Distributed Systems: Principles and Paradigms*. Prentice-Hall.

CBE312 Internet of Things and its Security [3-0-2-4]

Course Prerequisites

Student should have a passing grade in Computer Networks (ICS 224), Digital Design and Electric Circuits (IEC 121), Computer Programming (ICS 112) or the instructor's approval.

Course Objectives

- Introduce the concept and the basics of Wireless, IoT and Cloud technologies.
- Analyze various secured Wireless Communication Protocols for IoT Infrastructure.
- Provide knowledge on various applications of IoT-based technologies and their associated circuits.

- Enable awareness on the different IoT vulnerabilities, attacks, and security methods.

Course Outcomes

Students who successfully complete this course will be able to:

- Learn the basics of communication in wireless sensor networks and cloud computing.
- Compare various secured wireless communication protocols for IoT infrastructure.
- Understand the various applications of IoT.
- Design IoT-based applications using Arduino or Raspberry Pi boards.
- Understand various attacks and different security measures in IoT infrastructure.

Syllabus

Introduction: Basics of networking - wired, wireless, MANET, PAN, Wireless Sensor Networks, M2M Communication.

Secured Wireless Communication Protocols for IoT Infrastructure: IPv6 -LowPAN, LoRa, Transport-Bluetooth-LPWAN, Data -MQTT -CoAP.

IoT Architectures and Programming: Basic architectures, Sensor basics, sensing and actuation, sensor communications, connectivity challenges, data processing mechanisms, scalability issues, visualization issues, analytics basics, utility of cloud computing, fog computing, and edge computing, advanced IoT architectures, Raspberry Pi and Arduino programming.

Applications: IoT for industrial automation (Industry 4.0), smart city, smart home, smart transportation, smart healthcare, smart agriculture, Golang-based implementation.

IoT Security: Vulnerabilities, attacks, and countermeasures, security engineering for IoT development, IoT security lifecycle.

Textbooks/References

1. Pethuru Raj and Anupama C. Raman, *The Internet of Things: Enabling Technologies, Platforms, and Use Cases*, CRC Press, First edition, 2017.
2. B. Russell and D. Van Duren, *Practical Internet of Things Security*, Packt Publishing, 2016.
3. Fei Hu, *Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations*, CRC Press, 2016.
4. Honbu Zhou, *The Internet of Things in the Cloud: A Middleware Perspective*, CRC Press, First edition, 2012.
5. Arshdeep Bahga and Vijay Madisetti, *Internet of Things: A Hands-on Approach*, Universities Press, First edition, 2014.

6. Mung Chiang, Bharath Balasubramanian, Flavio Bonomi, *Fog for 5G and IoT* (Information and Communication Technology Series), Wiley Series, First edition, 2017.
7. Alan A. A. Donovan, Brian W. Kernighan, *The Go Programming Language*, Addison-Wesley Professional C

CBS311 Fundamentals of Data Science [3-0-2-4]

Course Prerequisites

Student should have a passing grade in Calculus and Linear Algebra (IMA 121), Probability, Statistics and Random Processes (IMA 211), and Data Structures I (ICS 121) or the instructor's approval.

Course Objectives

- To introduce the concepts, techniques, and methodologies in data science.
- To provide skills required to uncover patterns and underlying relationships in small data.
- To understand the principles of optimization.
- To introduce classical data mining techniques and related tools.
- To introduce a data analytics problem-solving framework.

Course Outcomes

Students who successfully complete this course will be able to:

- Comprehend the various phases involved in the data science process.
- Apprehend the differences between related tasks in data science.
- Perform data pre-processing and data visualization.
- Design and develop classification systems for real-world problems.
- Perform clustering analysis.

Syllabus

Data Pre-processing and Cleaning: Overview of Data Science, Data preparation, Data pre-processing, Data cleaning (handling missing data, outliers), Data integration and transformation, Feature engineering and extraction, Data sampling, Data reduction, Data discretization, Outlier detection, Knowledge mining from databases, Similarity measures.

Data Analytics and Visualization: Data summarization, Exploratory Data Analysis (EDA), tools and techniques, Application of transforms for data analysis.

Optimization and Machine Learning Fundamentals: Overview of multivariate optimization problems (linear, non-linear, constrained, unconstrained), Gradient and search-based optimization for machine learning, Typology of data science problems and a solution framework.

Regression Analysis: Simple linear regression, Multivariate linear regression, Polynomial regression.

Classification: Logistic regression, Decision trees and random forests, SVMs.

Clustering and Anomaly Detection: Clustering methods, Outlier detection.

Dimension Reduction: Principal Component Analysis (PCA).

Lab Practice: Manipulating and visualizing data with R/Python, Outlier detection, Statistical analysis on data, Data wrangling and cleaning, Probability distributions, Analysis of variance, Feature engineering and extraction, Data mining ML algorithms: Classification, Clustering.

Textbooks/References

1. Kotu, V. and Deshpande, B. (2018) *Data Science: Concepts and Practice*, Morgan Kaufmann.
2. Saltz, J.S. and Stanton, J.M. (2017) *An Introduction to Data Science*, SAGE Publications.
3. Kelleher, J.D., and Tierney, B. (2018) *Data Science*, MIT Press.
4. Jiawei Han. *Data Mining: Concepts and Techniques*, Morgan Kaufmann Publishers.
5. Tan, Pang-Ning, Michael Steinbach, and Vipin Kumar. *Introduction to Data Mining*, Pearson Education India, 2016.
6. Ricci, Francesco, Lior Rokach, and Bracha Shapira. *Introduction to Recommender Systems Handbook*, Springer, Boston, MA, 2011.
7. Witten, E. Frank, M. Hall. *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann Publishers, 2011.
8. Garrett Golemund, Hadley Wickham. *R for Data Science*, O'Reilly Media Inc., 2016.
9. Cichosz, Pawel. *Data Mining Algorithms: Explained Using R*, John Wiley & Sons, 2014.
10. Raghunathan Rengaswamy, Resmi Suresh, *Data Science for Engineers*, CRC Press, 1st edition, 2022.

IMA313 Number Theory and Mathematical Theory of Coding [3-0-0-3]

Course Objectives

- To understand the basics of number theory, theory of congruences, and modular arithmetic and to apply the same in data security.
- To analyze various error correction codes.

Course Outcomes

Students who successfully complete this course will be able to:

- Apply the basics of number theory in different scenarios.
- Learn different error-correcting codes.

Syllabus

Introduction to Number Theory: Theory of congruences and applications, Divisibility theory - division algorithm, the greatest common divisor, the Euclidean algorithm, the Diophantine equation, Primes and their distribution, The fundamental theorem of arithmetic (no proof) and applications.

The Theory of Congruences: Binary and decimal representation of integers, Congruences in one unknown, Chinese remainder theorem, Fermat's theorem, Pseudo primes, Fermat factorization.

Number Theoretic Functions: Numbers of special form, Number theoretic functions - the sum and number of divisors, the greatest integer function, an application to the calendar, Euler's phi function, Euler's theorem.

Numbers of Special Form: Perfect numbers, Fermat numbers, Legendre symbol, Jacobi symbol.

Algebraic Structures: Finite fields, The order of integer modulo, Primitive roots, Discrete logarithms, Quadratic residue, Quadratic reciprocity law, Primality testing.

Introduction to Groups, Rings, and Fields: Polynomial arithmetic, Finite fields.

Introduction to Coding Theory: Error-correcting codes, Basic concepts and definitions, Encoding and decoding, Bounds on general codes, Linear codes, Syndrome decoding, The dual code, Hamming codes, Cyclic codes, Reed-Solomon codes.

Textbooks/References

1. D. A. Burton, *Elementary Number Theory*, 6/e, Tata McGraw Hill, 2007.
2. Stallings W., *Cryptography and Network Security: Principles and Practice*, 4/e, Pearson Education Asia, 2006.
3. Wade Trappe, Lawrence C. Washington, *Introduction to Cryptography with Coding Theory*, Second Edition.
4. Niven I., Zuckerman H. S., and Montgomery H. L., *An Introduction to Theory of Numbers*, 5/e, John Wiley and Sons, 2004.
5. Joseph A. Gallian, *Contemporary Abstract Algebra*, Narosa, 1998.

IHS314 Financial Crime, Motivations and Typologies [3-0-0-3]

Course Objectives

- To understand the tracing of illicit financial transactions and various anti-money laundering laws.
- To introduce the fundamental concepts, structures, and dynamics of online social networks.
- To study various methods and techniques for collecting, processing, and analyzing data from online social networks.
- To explore the concepts of information privacy, disclosure, and their effects in online social networks.
- To introduce the basic concepts of phishing and fraud detection.

Course Outcomes

At the end of the course, candidates should be able to:

- Understand tracing illicit financial transactions.
- Understand the various anti-money laundering laws.
- Understand online social networks, data collection techniques, trust, credibility, and security issues in social systems, as well as strategies for countering threats.
- Understand various strategies and best practices for preventing phishing attacks.
- Understand the signs and behaviors that may indicate fraudulent accounts, profiles, or activities.

Syllabus

Tracing Illicit Transactions: Identify the common areas of interview questions in tracing financial crime evidence. State and explain the main sources of income and expenditure in illicit financial crime. Distinguish between the direct and indirect methods of tracing illicit financial transactions. Describe the investigation procedures for tax fraud. Describe the indirect method of tracing financial crime. Explain the Net Worth Method of investigating financial crime. Prepare a profile of a financial fraudster. Determine the net worth of a financial crime suspect.

Anti-Money Laundering Laws: Define money laundering and identify the process of money laundering. Identify the major money laundering countries in the world. Explain money laundering and describe the penalty for money laundering under the Money Laundering Act. Describe the composition of the Financial Intelligence Centre. Define accountable institutions and state their composition. Identify the various international bodies that impose money laundering sanctions. Explain the various types of money laundering sanctions.

Foundations of Online Social Networks and Data: Introduction to online social networks, data collection, and API basics. Understanding the essence of online social networks, data gathering techniques, and the role of APIs. Exploring effective methods for collecting data from online social media.

Trust, Credibility, and Security: Analyzing the concepts underpinning trust and credibility within social systems. Investigating the role of online social media in law enforcement and security measures. Examining privacy implications, disclosure, and their impact on online social media and networks.

Counteracting Threats: Learning to identify and combat phishing in online social media while uncovering techniques for spotting fraudulent entities within online social networks.

Textbooks/References

1. Tracy L. Coenen, *Expert Fraud Investigation: A Step-by-Step Guide*, 2020.
2. *ACFE Fraud Examiners Manual*.
3. George A. Manning, *Financial Investigation and Forensic Accounting*, Routledge, 2010.
4. Arun Srivastava and Andrew Keltie, *A Practitioner's Guide to the Law and Regulation of Financial Crime*, 2010.
5. Yuval Elovici, Yaniv Altshuler, *Security and Privacy in Social Networks*, Springer-Verlag New York, 2019.

CBS312 Introduction to Artificial Intelligence [1-0-0-1]

Course Prerequisites

Student should have a passing grade in Data Structures I (ICS 121), Design and Analysis of Algorithms (ICS 211), or the instructor's approval.

Course Objectives

- To provide a historical perspective and broad introduction to Artificial Intelligence (AI).
- To introduce the fundamental concepts and principles of AI for problem-solving, knowledge representation, learning, and inference.
- To provide a solid grounding in fuzzy logic and related concepts.
- To impart knowledge on expert systems tools and applications.

Course Outcomes

- Demonstrate in-depth understanding of the fundamentals of AI.
- Ability to solve basic problems by applying concepts of problem-solving, knowledge representation, learning, and inference.
- Design solutions for optimization problems using fuzzy logic or genetic algorithms.
- Apply AI techniques to develop expert systems.

Syllabus

Evolution and Fundamentals of AI: History of Artificial Intelligence (AI) and its applications, Agents, Problem Spaces and Search - Uninformed and informed search, Problem solving - Adversarial search and games, Search spaces - Weak methods, game trees.

Knowledge Representation and Learning: Propositional logic, Predicate logic, Constraint-satisfaction problems.

Uncertainty Reasoning: Bayesian Networks, Fuzzy logic - Membership functions, Fuzzy inference systems, Fuzzy knowledge and rule-based system.

Expert Systems: Overview of expert systems, Inference, Forward chaining and backward chaining.

Textbooks/References

1. Stuart Russell, Peter Norvig, *Artificial Intelligence: A Modern Approach*, Prentice Hall, Fourth Edition, 2020.
2. Luger, George F., *Artificial Intelligence: Structures and Strategies for Complex Problem Solving*, Pearson Education, 2005.
3. Nils J. Nilsson, *Artificial Intelligence: A New Synthesis*, Morgan-Kaufmann, 1998.
4. Elaine Rich, Kevin Knight, *Artificial Intelligence*, McGraw-Hill, Second Edition, 1991.
5. Timothy J. Ross, *Fuzzy Logic with Engineering Applications*, John Wiley and Sons, Third Edition, 2011.

SEMESTER VI

CBE321 Machine Learning and Cyber Security [3-0-0-3]

Course Objectives

- To provide basic understanding of core concepts of machine learning and its process.
- To provide an in-depth introduction to supervised, unsupervised learning and its application to specific security problems.
- To design and implement machine learning solutions for various security applications.

Course Outcomes

- Understand machine learning and its benefits in specific security problems.
- Learn machine learning techniques such as supervised and unsupervised algorithms.
- Learn the use of machine learning for detecting and mitigating cyber threats in various applications.
- Develop an ability to select appropriate machine learning algorithms for specific security applications to efficiently tackle security problems.

Syllabus

Machine Learning Overview: Introduction, Linear Regression, Maximum Likelihood Estimation (MLE), Bias/Variance trade-offs, Technical requirements (Data pre-processing, Standardization, Dimension reduction, Feature selection, Train-Test splitting, Loss function, Optimization, Model selection, Cross-validation).

Supervised Learning: Naïve Bayes classifier, Support Vector Machine (SVM), Regularization, Classification errors, Decision Tree, K-Nearest Neighbors (KNN), Artificial Neural Network (ANN).

Unsupervised Learning: Clustering (K-means, K-medoids, Hierarchical clustering algorithms, DBSCAN).

Machine Learning for Cyber Security: Machine Learning for Malware detection, Supervised Learning for Misuse/Signature Detection, Anomaly Detection using ML, Spam detection based on Machine Learning approach, Adversarial Machine Learning.

Textbooks/References

1. Mitchell, Tom, *Machine Learning*, McGraw-Hill, 1997.
2. Bishop, C. M., *Pattern Recognition and Machine Learning*, Springer, 2006.
3. Langley, P., *Elements of Machine Learning*, Morgan Kaufmann, 1995.

4. Hastie, T., R. Tibshirani, and J. H. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, Second Edition, Springer, 2009.
5. Dua, Sumeet, and Xian Du, *Data Mining and Machine Learning in Cybersecurity*, CRC Press, 2016.
6. Chio, Clarence, and David Freeman, *Machine Learning and Security: Protecting Systems with Data and Algorithms*, O'Reilly Media, 2018.

CBE322 Digital Forensics [3-0-2-4]

Course Prerequisites

Student should have a passing grade in IT Workshop I (ICS111) and Operating Systems (ICS 212) or the instructor's approval.

Course Objectives

- Summarize the basic principles of computer forensics.
- Understand various stages of digital forensics investigation.
- Summarize important laws relevant to digital forensics investigation.
- Describe the digital forensic lab and tools.
- Understand storage management and its importance in forensics.
- Understand the file system fundamentals and the internals of Windows and Linux file systems.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand the principles and stages of computer forensics.
- Comprehend various legal aspects pertaining to digital forensic investigation.
- Understand various Windows and Linux file systems and their operations.
- Grasp digital forensic tools and the setup of a digital forensic lab.

Syllabus

Introduction to Computer Forensics: Foundations of Digital Forensics, Need for Digital Forensics, Forensic Investigation Model.

Computer Hardware and Operating Systems: Overview of computer hardware and operating systems, Storage media/devices and their working, Windows and Linux File Systems.

Forensic Investigation: First Responder, Forensic data acquisition, Forensic data validation, Evidence Collection and Analysis Techniques, Disk Wiping.

Legal Issues: Computer crimes and legal aspects of digital forensics.

Textbooks/References

1. Bill Nelson, Amelia Phillips, Christopher Steuart, *Guide to Computer Forensics and Investigations*, 6th Edition, Cengage Learning.
2. Chuck Easttom, *System Forensics Investigation and Response*, 3rd Edition, Jones & Bartlett Learning.
3. Cory Altheide, Bruce Nikkel, Harlan Carvey, *Digital Forensics with Open Source Tools*, Elsevier.
4. Marjie T. Britz, *Computer Forensics and Cyber Crime: An Introduction*, Pearson.
5. Cory Altheide, Bruce Nikkel, Harlan Carvey, *Digital Forensics with Open Source Tools*, Elsevier Science & Technology.

CBS321 Cloud Computing and Security [3-0-2-4]

Course Objectives

- Understand fundamental cloud computing concepts, including service and deployment models.
- Identify security challenges in cloud environments and mitigation strategies.
- Explore data lifecycle management and protection mechanisms in the cloud.
- Learn identity and access management principles for secure cloud operations.
- Gain hands-on experience with cloud security tools across AWS, Azure, and Google Cloud.

Course Outcomes

Students who successfully complete this course will be able to:

- Explain cloud computing concepts and their security implications.
- Implement data protection strategies and identity management in the cloud.
- Utilize AWS, Azure, and Google Cloud security services for threat detection and compliance.
- Conduct cloud security audits and enforce security policies.
- Design and implement secure cloud solutions for enterprise environments.

Syllabus

Fundamentals of Cloud Computing: Cloud Deployment Models, Cloud Computing Roles and Risks.

Key management, Security Challenges – Data Lifecycle – Data Protection – Identity life cycle X.509 certificates, OpenSSL.

Data audit: EBS and S3 -KMS -AWS Security Hub-AWS Guard Duty -AWS Shield -AWS Firewall Manager- AWS WAF - AWS Macie - AWS Config - AWS Elastic Load Balancing (ELB) -Cloud watch - Cloud Trail - Identity and Access Management. Microsoft Defender for Cloud-Azure DDoS Protection-Azure Firewall-Azure Information Protection- Azure Security Center-Azure Front Door- WAF-Azure Active Directory-Azure Policy-Azure Key Vault
Google Cloud Security Command Center -Chronicle Security Operations-Google Cloud Armor- Google Cloud Firewall -Google Cloud Data Loss Prevention (DLP)-Google Cloud CDN - Cloud Load Balancing-Google Cloud Identity and Access Management

Textbooks/References

1. Anthony T. Velte, Toby J. Velte, Robert Elsenpeter, *Cloud Computing: A Practical Approach*, Tata McGraw-Hill, 2010.
2. Michael Miller, *Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online*, Que, 2008.
3. Tim Mather, Subra Kumaraswamy, Shahed Latif, *Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance*, O'Reilly Media, 2009.
4. Ronald L. Krutz, Russell Dean Vines, *Cloud Security*, Wiley, 2010.
5. John Rittinghouse, James Ransome, *Cloud Computing*, CRC Press, 2009.
6. J.R. ("Vic") Winkler, *Securing the Cloud*, Syngress, 2011.
7. Cloud Security Alliance, *Security Guidance for Critical Areas of Focus in Cloud Computing*, 2009.
8. VMware, *VMware Security Hardening Guide*, White Paper, 2011.
9. Cloud Security Alliance 2010, 2024, *Top Threats to Cloud Computing*, Microsoft, 2024.
10. Timothy Grance, Wayne Jansen, NIST, *Guidelines on Security and Privacy in Public Cloud Computing*, 2011.
11. <https://cloud.google.com/docs>
12. <https://docs.aws.amazon.com/>
13. <https://learn.microsoft.com/en-us/azure/?product=popular>

CBE323 Cryptography [3-0-2-4]

Course Prerequisites

Student should have a passing grade in Number Theory and Mathematical Theory of Coding (IMA 312), Discrete Mathematics (IMA 111).

Course Objectives

- To lay a foundation on Security in Networks, Classical Cryptosystem, and Block Cipher Modes of Operation.

- To analyse various Private and Public key Cryptosystem for encryption, key exchange, hashing, and authentication protocols.
- To acquire fundamental knowledge on applications of cryptography.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand the fundamental concepts of Classical and modern Cryptosystem.
- Compare various private and public key Cryptosystems for encryption, key exchange, and authentication algorithms.
- Understand the different applications of cryptography.

Syllabus

Introduction: Cryptography, cryptanalysis, cryptology, Classical Cryptosystem - Shift cipher, Affine cipher, Vigenère cipher, Substitution, Transposition techniques.

Block Ciphers and Modes of Operations: Data Encryption Standard (DES) - Block cipher principles - Block cipher modes of operation, AES - Triple DES - Blowfish - RC5.

Public Key Cryptography: Public Key Cryptosystem, Key distribution, Diffie-Hellman Key Exchange, Man-in-the-Middle (MITM) Attack, RSA, Random Number Generation, Elliptic Curve Cryptography (ECC), Key Management.

Hash Functions and Digital Signatures: Authentication requirement, Authentication function, Message Authentication Code (MAC), Hash functions, SHA, HMAC, Digital signature and authentication protocols.

Applications: Authentication – Kerberos, IP Security – IPSec, Web Security - SSL, TLS, Blockchain, IoT Security.

Textbooks/References

1. William Stallings, *Cryptography and Network Security*, 6th Edition, Pearson Education.
2. Behrouz A. Forouzan, Debdeep Mukhopadhyay, *Cryptography and Network Security*, 5th Edition, McGraw Hill Education.
3. Rich Helton, Johennie Helton, *Mastering Java Security: Cryptography Algorithms and Practices*, John Wiley Publishers.
4. Charles P. Pleege, *Security in Computing*, 5th Edition, Pearson Education Asia.
5. William Stallings, *Network Security Essentials: Applications and Standards*, Pearson Education Asia.
6. Charlie Kaufman, Radia Perlman, Mike Speciner, *Network Security: Private Communication in a Public World*, 6th Edition, Prentice Hall India.

ISC322 Criminal Psychology and Behavior Intelligence [3-0-0-3]

Course Objectives

- To make the students familiar with the field of Criminal Psychology.
- To make the students understand the origins of Criminal Behaviour.

Course Outcomes

Students who successfully complete this course should have a comprehensive understanding of Criminal Behaviour and Psychology.

Syllabus

Nature and History of Criminal and Forensic Psychology: Social context of Crime, Extent of Criminality, Changing nature of Crime, Conservative and Radical interpretations in complexity of victimization.

Types of Offenders: Violent Offenders, Media influences and Research Statistics, Theories of Homicide, Psychological disposition, Socio-Biological theory, Multi-Factorial Approach.

Mental Illness and Crime: Problem of evidence, Mental illness and Crime in general.

Eyewitness Testimony: Accuracy of witness evidence in Court, Witness confidence and improving the validity of line-up, Clinical approaches in Risk and danger assessment.

Textbooks/References

1. Dennis Howitt, *Introduction to Forensic and Criminal Psychology*, 6th Edition, Pearson, 2018.
2. Wayne Petherick, Brent Turvey, Claire Ferguson, *Forensic Criminology*, 1st Edition, Elsevier, ISBN: 9780123750716.
3. Bruce Arrigo, Stacey Shipley, *Introduction to Forensic Psychology*, 2nd Edition, Academic Press, ISBN: 9780080468532.

IOE322 Information Security Standards, Policies, Strategies & Audits [3-0-0-3]

Course Objectives

- Enable a clear understanding and knowledge of Security Analyst foundations.
- Expose students to various IT auditing techniques.
- Understand the significance of Risk Management.

Course Outcomes

Students who successfully complete this course should have a comprehensive understanding of Information Security Standards, auditing process, and Risk Management.

Syllabus

Introduction and IT Audit: IT Environment, Methods for Business Advisory Audits, Role of the IT Audit Team, IT Audit Process, Identifying what to Audit, Stages of Auditing.

Auditing Techniques: Auditing Entity-Level Controls, Auditing Cybersecurity Programs, Auditing Data Centers and Disaster Recovery, Auditing Networking Devices, Auditing Windows and Linux Operating Systems, Auditing Web Servers and Web Applications, Auditing Databases, Auditing Storage, Auditing Virtualized Environments, Auditing End-User Computing Devices, Auditing Applications, Auditing Company Projects.

Frameworks, Standards, Regulations, and Risk Management: Internal IT controls, Benefits of Risk Management, Quantitative Risk Analysis, Qualitative Risk Analysis.

Textbooks/References

1. Mike Kegerreis, Mike Schiller, Chris Davis, *IT Auditing Using Controls to Protect Information Assets*, 3rd Edition, McGraw-Hill Education, 2019, ISBN-10: 1260453227.
2. Angel R. Otero, *Information Technology Control and Audit*, 5th Edition, Auerbach Publications, 2020, ISBN-10: 1498752284.
3. Martin Weiss, Michael G. Solomon, *Auditing IT Infrastructures for Compliance*, 2nd Edition, Jones & Bartlett Learning, 2015, ISBN-10: 1284090701.
4. Stephen D. Gantz, *The Basics of IT Audit: Purposes, Processes, and Practical Information*, Syngress, 2013, ISBN-10: 0124171591.

SEMESTER VII

CBS411 Mobile Forensics and Security [3-0-2-4]

Course Objectives

- To gain knowledge on mobile phone evidence extraction process.
- To understand the practical mobile forensic approaches.
- To engage students in forensic acquisition and analysis of mobile computing devices, specifically Android devices.
- To gain an understanding of mobile device identification.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand what data can be acquired from mobile devices and be able to acquire and investigate data from mobile devices using forensically sound and industry-standard tools.
- Comprehend the relationship between mobile and desktop devices in relation to criminal and corporate investigations.
- Analyse mobile devices, their backup files, and artifacts for forensic evidence.

Syllabus

Introduction to Mobile Forensics: Need for mobile forensics, Challenges in mobile forensics.

The Mobile Phone Evidence Extraction: Documenting and reporting phase, Presentation phase, Archiving phase, Practical mobile forensic approaches: Overview of mobile operating systems, Data acquisition methods, Examination and analysis of evidence stored on mobile phones.

Understanding Android: Android model, Android security - Secure kernel, Security-Enhanced Linux, Full Disk Encryption, Trusted Execution Environment, Android file system.

Android Forensic Techniques: Android forensic setup and pre-data extraction techniques, Android data extraction techniques, Android data analysis and recovery, Android data recovery.

Android App Analysis, Malware, and Reverse Engineering: Analyzing Android apps, Reverse engineering Android apps, Extracting an APK file from an Android device, Android malware.

Textbooks/References

1. Bommisetty, Satish, Rohit Tamma, and Heather Mahalik, *Practical Mobile Forensics*, Packt Publishing Ltd, 2014.
2. Tamma, Rohit, and Donnie Tindall, *Learning Android Forensics*, Packt Publishing Ltd, 2015.

3. Angus M. Marshall, *Digital Forensics: Digital Evidence in Criminal Investigation*, John Wiley & Sons, 2008.
4. Mikhaylov, Igor, *Mobile Forensics Cookbook: Data Acquisition, Extraction, Recovery Techniques, and Investigations Using Modern Forensic Tools*, Packt Publishing Ltd, 2017.
5. Joakim, K., *Fundamentals of Digital Forensics: Theory, Methods, and Real-Life Applications*, Springer International Publishing, Berlin, Heidelberg, 2018.

CBE411 Vulnerability Assessment and Penetration Testing [3-0-2-4]

Course Prerequisites

Student should have a passing grade in Computer Networks (ICS 224), Database Security (CBS 311), and Network Security, IoT and Wireless Security (CBS 312), or the instructor's approval.

Course Objectives

- Introduces the concepts of Penetration testing.
- Gives the students the opportunity to learn about different tools and techniques for penetration testing and security.
- Practically apply penetration testing tools to perform various activities.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand the core concepts related to vulnerabilities and their causes.
- Understand ethics behind hacking and vulnerability disclosure.
- Comprehend the impact of hacking.
- Exploit the vulnerabilities related to computer systems and networks using state-of-the-art tools and technologies.

Syllabus

Introduction and Information Security Overview: Hacking and Ethical Hacking concepts, Hacker behavior & mindset, Hacking methodology - MITRE ATT&CK framework, Cyber Kill Chain

Reconnaissance and foot printing, passive reconnaissance, Active reconnaissance Methodology- IP spoofing, DNS spoofing. Active scanning, Enumerating Common services, Penetration Testing Reports.

Attacks and Countermeasures: Social engineering attacks and countermeasures, Phishing, Password attacks, Techniques for wordlist Creation, Privilege escalation, Network infrastructure vulnerabilities, IP spoofing, DNS spoofing. Web server and application vulnerabilities, DoS attacks.

Penetration Testing Tools: Metasploit framework, Penetration testing tools in Kali Linux.

Textbooks/References

1. Baloch, R., *Ethical Hacking and Penetration Testing Guide*, Auerbach Publications, CRC Press, 2015.
2. David Kennedy, Jim O'Gorman, Devon Kearns, Mati Aharoni, *Metasploit: The Penetration Tester's Guide*, No Starch Press, 2011, ISBN: 159327288X, 9781593272883.
3. Sagar Rahalkar, *Quick Start Guide to Penetration Testing: With NMAP, OpenVAS and Metasploit*, 1st Edition, Apress Publications, 2019, Softcover ISBN: 978-1-4842-4269-8.
4. Christopher Hadnagy, *Social Engineering: The Science of Human Hacking*, 2nd Edition, Wiley Publisher, 2018, ISBN-13: 978-1119433385.
5. Glen D. Singh, *Learn Kali Linux 2019: Perform Powerful Penetration Testing Using Kali Linux, Metasploit, Nessus, Nmap, and Wireshark*, Packt Publishing, 2019, ISBN: 1789611806.
6. Michael Hixon, Justin Hutchens, *Kali Linux: Network Scanning Cookbook*, Packt Publishing, 2017, ISBN: 139781787287907.

CBE412 Multimedia Security & Forensics [3-0-2-4]

Course Objectives

- Introduce multimedia, its application areas, data encoding, and compression techniques.
- Develop an understanding of Quality of Service and its constraints.
- Understand synchronization concepts.
- Discuss multimedia security attacks and defense techniques.
- Introduce multimedia forensic concepts.

Course Outcomes

Students who successfully complete this course will be able to:

- Demonstrate how quality of service can be ensured in multimedia applications.
- Apply different synchronization techniques on multimedia.
- Ensure security of multimedia applications.
- Perform forensic analysis on multimedia data.

Syllabus

Introduction: Multimedia application areas, Interdisciplinary aspects of multimedia, Multimedia data encoding, Concept of data compression in the multimedia field.

Quality of Service & Operating System: Requirements and constraints, Quality of Service concept, Resource management, Media server architecture, Storage management, Services, Protocols, Layers.

Security in Multimedia Applications: Security attacks, Multimedia encryption, Steganography, Digital image watermarking, Multimedia authentications.

Multimedia Evidence Handling: Digital forensics laboratories in operation, Standards and best practices in digital and multimedia forensics, Digital evidence extraction, Multimedia file carving, Multimedia device and source forensics: Forensic camera model identification, Printer and scanner forensics, Microphone forensics, Multimedia content forensics.

Textbooks/References

1. Ralf Steinmetz, Klara Nahrstedt, *Multimedia Systems*, Springer International Edition.
2. Ho, Anthony TS, and Shujun Li, eds., *Handbook of Digital Forensics of Multimedia Data and Devices*, John Wiley & Sons, 2015.
3. John. F. Koegel Buford, *Multimedia Systems*, Pearson Education.
4. Ingemar Cox, Matthew Miller, Jeffrey Bloom, Jessica, *Digital Watermarking and Steganography*.
5. Steinmetz, Ralf, Jana Dittmann, and Martin Steinebach, eds., *Communications and Multimedia Security Issues of the New Century: IFIP TC6/TC11 Fifth Joint Working Conference on Communications and Multimedia Security (CMS'01) May 21–22, 2001, Darmstadt, Germany*, Vol. 64, Springer, 2013.

IOE411 Blockchain & Crypto-Currencies [3-0-0-3]

Course Prerequisites

The participants should have prior knowledge of the following topics/courses:

- Computer Networks
- Cryptography
- Linux commands

Course Objectives

- Introduce the concept and basics of blockchain technologies.
- Enable awareness of the different generations of blockchains.
- Provide knowledge on various applications of blockchain technologies.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand the basics of blockchain technologies and their various applications.
- Understand the cryptocurrency Bitcoin.

- Implement smart contracts using Ethereum.
- Identify problems on which blockchains could be applied.

Syllabus

Introduction: Blockchain history, basics, architectures, Types of blockchain, Base technologies – Dockers, Hash function, Digital Signature - ECDSA, Zero Knowledge Proof.

Bitcoins: Fundamentals, aspects of bitcoins, properties of bitcoins, bitcoin transactions, bitcoin P2P networks, block generation in bitcoins, consensus algorithms - Proof of Work, Proof of Stake, Proof of Burn.

Ethereum: Introduction to Ethereum, Consensus Mechanism, Implementation, Transactions, demonstrations, Smart Contracts.

Applications: Blockchain applications, e-governance, smart cities, smart industries, anomaly detections, use cases, trends in blockchains, serverless blocks, scalability issues, blockchain on clouds.

Textbooks/References

1. Baxv Kevin Werbach, *The Blockchain and the New Architecture of Trust*, MIT Press, 2018.
2. Joseph J. Bambara and Paul R. Allen, *Blockchain – A Practical Guide to Developing Business, Law, and Technology Solutions*, McGraw Hill, 2018.
3. Joseph J. Bambara and Paul R. Allen, *Blockchain, IoT, and AI: Using the Power of Three to Develop Business, Technical, and Legal Solutions*, Barnes & Noble Publishers, 2018.
4. Melanie Swan, *Blockchain – Blueprint for a New Economy*, O'Reilly Publishers, 2018.
5. Jai Singh Arun, Jerry Cuomo, Nitin Gaur, *Blockchain for Business*, Pearson Publishers, 2019.
6. Satoshi Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*.

SEMESTER VIII

CBS421 Cyber Ethics, Privacy and Legal Issues [3-0-0-3]

Course Objectives

- Deals with all aspects of Cyber Law as per the Indian IT Act.
- Covers an overview of Cyber Ethics, Intellectual Property Rights, and Trademark-related laws with respect to cyberspace.

Course Outcomes

Students who successfully complete this course will be able to:

- Demonstrate a critical understanding of Cyber Law with respect to the Indian IT Act, Cyber Ethics, and Intellectual Property Rights.

Syllabus

Cyber Crimes: Categories and kinds, Evolution of the IT Act, IT Act, 2000, various authorities under the IT Act and their powers, Penalties & Offences, amendments.

Case Laws on Cyber Space: Jurisdiction and jurisdiction issues under the IT Act, E-commerce and laws in India, Digital/Electronic Signature in Indian Laws.

Intellectual Property Rights: Domain names and trademark disputes, Copyright in computer programs, Concept of patent rights, Sensitive personal data or information in Cyber Law, Cyber Law from an international perspective.

Cyber Ethics and Code: Net neutrality, Free speech and censorship in cyberspace, Intellectual Property in cyberspace, Privacy rights and surveillance.

Textbooks/References

1. Sushma Arora, Raman Arora, *Cyber Crimes & Laws*, 4th Edition, Taxmann, 2021, ISBN-10: 9390712491.
2. N S Nappinai, *Technology Laws Decoded*, 1st Edition, Lexis Nexis, ISBN: 9789350359723.
3. Suresh T. Vishwanathan, *The Indian Cyber Law*, Bharat Law House, New Delhi.
4. P.M. Bukshi and R.K. Suri, *Guide to Cyber and E-Commerce Laws*, Bharat Law House, New Delhi.
5. Rodney D. Ryder, *Guide to Cyber Laws*, Wadhwa and Company, Nagpur.
6. *The Information Technology Act, 2000*, Bare Act – Professional Book Publishers, New Delhi.
7. Richard A. Spinello, *Cyberethics: Morality and Law in Cyberspace*, 7th Edition, Jones & Bartlett Learning, 2020, ISBN-10: 1284184064.

CBS422 Biometric Security [3-0-2-4]

Course Objectives

- To understand the technologies of fingerprint, iris, face, and speech recognition.
- To understand the general principles of the design of biometric systems and the underlying trade-offs.
- To recognize personal privacy and security implications of biometrics-based identification technology.
- To identify issues in the realistic evaluation of biometrics-based systems.

Course Outcomes

Students who successfully complete this course will be able to:

- Demonstrate knowledge of the basic physical, biological science, and engineering principles underlying biometric systems.
- Understand and analyze biometric systems at the component level and be able to analyze and design basic biometric system applications.
- Demonstrate knowledge of engineering principles underlying biometric systems.
- Analyze and design basic biometric system applications.
- Understand various biometric security issues.

Syllabus

Biometrics: Introduction, Benefits of biometrics over traditional authentication systems, Benefits of biometrics in identification systems, Selecting a biometric for a system, Applications, Key biometric terms and processes, Biometric matching methods, Accuracy in biometric systems.

Fingerprint Identification Technology: Fingerprint Patterns, Fingerprint Features, Fingerprint Image, Width between two ridges, Fingerprint Image Processing, Minutiae Determination, Fingerprint Matching: Fingerprint Classification, Matching Policies.

Face Recognition: Introduction, Components, Facial Scan Technologies, Face Detection, Face Recognition, Representation and Classification, Kernel-based Methods and 3D Models, Learning the Face Space, Facial Scan Strengths and Weaknesses, Methods for Assessing Progress in Face Recognition.

Fusion in Biometrics: Introduction to Multibiometric Systems, Information Fusion in Biometrics, Issues in Designing a Multibiometric System, Sources of Multiple Evidence, Levels of Fusion in Biometrics – Sensor Level, Feature Level, Rank Level, Decision Level Fusion, Score Level Fusion.

Examples: Biopotential and gait-based biometric systems.

Case Study Presentations: Biometrics in the Banking Industry, Biometrics in Computerized Patient Records, Biometrics in Credit Cards, Biometrics in Mass Disaster Victim

Textbooks/References

1. James Wayman, Anil Jain, Davide Maltoni, Dario Maio, *Biometric Systems, Technology Design and Performance Evaluation*, Springer, 2005.
2. David D. Zhang, *Automated Biometrics: Technologies and Systems*, Kluwer Academic Publishers, New Delhi, 2000.
3. Arun A. Ross, Karthik Nandakumar, A.K. Jain, *Handbook of Multibiometrics*, Springer, New Delhi, 2006.
4. Samir Nanavathi, Michel Thieme, Raj Nanavathi, *Biometrics - Identity Verification in a Network*, 1st Edition, Wiley Eastern, 2002.
5. John Chirillo, Scott Blaul, *Implementing Biometric Security*, 1st Edition, Wiley Eastern Publication, 2005.
6. John Berger, *Biometrics for Network Security*, 1st Edition, Prentice Hall, 2004.
7. Paul Reid, *Biometrics for Network Security*, Pearson Education, 2004.
8. Nalini K Ratha, Ruud Bolle, *Automatic Fingerprint Recognition System*, Springer, 2003.
9. L. C. Jain, I. Hayashi, S. B. Lee, U. Halici, *Intelligent Biometric Techniques in Fingerprint and Face Recognition*, CRC Press, 1999.
10. John Chirillo, Scott Blaul, *Implementing Biometric Security*, John Wiley, 2003.
11. S.Y. Kung, S.H. Lin, M.W. Mak, *Biometric Authentication: A Machine Learning Approach*, Prentice Hall, 2005.

IOE422 Lightweight Cryptography [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Number Theory and Mathematical Theory of Coding (IMA 312), Discrete Mathematics (IMA 111) or the instructor's approval.

Course Objectives

- To familiarize with the design strategies and mathematics involved in Lightweight Cryptography.
- To analyze various lightweight cryptographic primitives.
- To identify the real-time applications of Lightweight Cryptography.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand the design strategies of lightweight ciphers to secure resource-constrained devices.
- Compare various lightweight cryptographic algorithms.

- Acquire the fundamental knowledge of the applications of lightweight cryptography.

Syllabus

Introduction: Overview of Lightweight Cryptography, Security Threats for resource-constrained devices, Design strategies for lightweight cryptography, Constraints and Compromises of Lightweight Algorithms, Modes of Operation.

Lightweight Cryptographic Primitives: Lightweight Block Ciphers: DESL and DESXL, PRESENT, CLEFIA, LED, SIMON and SPECK, TWINE, PRINCE, MIDORI, RECTANGLE, GRANULE, CRAFT. Lightweight Stream Ciphers, Lightweight HASH functions, Lightweight Message Authentication Codes.

Key Management and Applications of Lightweight Cryptography: Key management: Challenges in Designing IoT Group Key Management Protocols, Lightweight Group Key Management Protocols, Symmetric Key Constructions, Asymmetric Key Constructions. Applications: IPsec, TLS, Cyber-Physical Systems, IoT devices.

Textbooks/References

1. Kerry A. McKay, Larry Bassham, Meltem Sönmez Turan, Nicky Mouha, *Report on Lightweight Cryptography*, NIST, Computer Security Division Information Technology Laboratory, <https://doi.org/10.6028/NIST.IR.8114>.
2. Charalampos Maniavas, George Hatzivasilis, Konstantinos Fysarakis, Yannis Papaefstathiou, *A Survey of Lightweight Stream Ciphers for Embedded Systems*, 2015, <https://doi.org/10.1002/sec.1399>.
3. Bogdanov, A., Knudsen, L.R., Leander, G., Paar, C., Poschmann, A., *PRESENT: An Ultra-Lightweight Block Cipher*, CHES 2007, Springer, LNCS, 4727, pp. 450–466.
4. Shirai, T., Shibutani, K., Akishita, T., Moriai, S., Iwata, T., *The 128-bit Block Cipher CLEFIA*, FSE 2007, Springer, LNCS, 4593, pp. 181–195.
5. Gong, Z., Nikova, S., Law, Y.W., *KLEIN: A New Family of Lightweight Block Ciphers*, RFID Security and Privacy, Springer, LNCS 7055, pp. 1–18, 2012.
6. Guo, J., Peyrin, T., Poschmann, A., Robshaw, M., *The LED Block Cipher*, CHES 2011, Springer-Verlag, pp. 326–341.
7. Suzaki, T., Minematsu, K., Morioka, S., Kobayashi, E., *Twine: A Lightweight, Versatile Block Cipher*, ECRYPT Workshop on Lightweight Cryptography, 2011.
8. Engels, D., Saarinen, M.O., Schweitzer, P., Smith, E.M., *The Hummingbird-2 Lightweight Authenticated Encryption Algorithm*, RFID Security and Privacy, Springer, LNCS 7055, pp. 19–31, 2011.
9. Borghoff, J., et al., *PRINCE: A Low-latency Block Cipher for Pervasive Computing Applications*, ASIACRYPT 2012, Springer, LNCS, 7658, pp. 208–225.

10. Dai, Y., Chen, S., *Cryptanalysis of Full PRIDE Block Cipher*, Science China Information Sciences, 2014, [10.1007/s11432-015-5487-3](https://doi.org/10.1007/s11432-015-5487-3).
11. Albrecht, M.R., et al., *Block Ciphers Focus on the Linear Layer (feat. PRIDE)*, CRYPTO 2014, Springer, LNCS, vol. 8616, pp. 57–76.
12. Zhang, W., et al., *RECTANGLE: A Bit-slice Lightweight Block Cipher Suitable for Multiple Platforms*, SCIENCE CHINA Information Sciences, 2015, Vol. 58, 122103(15), [10.1007/s11432-015-5459-7](https://doi.org/10.1007/s11432-015-5459-7).
13. Banik, S., et al., *Midori: A Block Cipher for Low Energy*, Cryptology ePrint Archive, 2015, <https://eprint.iacr.org/2015/1142>.
14. Bansod, G., et al., *GRANULE: An Ultra Lightweight Cipher Design for Embedded Security*, IACR Cryptology ePrint Archive, 2018.
15. Beierle, C., et al., *CRAFT: Lightweight Tweakable Block Cipher with Efficient Protection Against DFA Attacks*, ToSC, 2019, <https://doi.org/10.13154/tosc.v2019.i1.5-45>.
16. Armknecht, F., et al., *A Guide to Fully Homomorphic Encryption*, IACR Cryptology ePrint Archive, 2015, 1192.
17. Boneh, Dan, Xavier Boyen, Eu-Jin Goh, *Hierarchical Identity-Based Encryption with Concerns and Challenges*, Egyptian Informatics Journal, 2020.
18. Mao, W., *Modern Cryptography: Theory and Practice*, Pearson Education, 2004.
19. Siddhartha, V., Gaba, G. S., Kansal, L., *A Lightweight Authentication Protocol using Implicit Certificates for Securing IoT Systems*, Procedia Computer Science, 2020, 167, 85–96, ELSEVIER.
20. Sun, W., et al., *Security and Privacy in the Medical Internet of Things: A Review*, Security and Communication Networks, 2018, Wiley.
21. Hammad, B.T., Jamil, N., Rusli, M.E., Z'aba, M.R., *A Survey of Lightweight Cryptographic Hash Functions*, International Journal of Scientific & Engineering Research, 2017.
22. Chowdhury, A., Dasbit, S., *LMAC: A Lightweight Message Authentication Code for Wireless Sensor Networks*, GLOBECOM 2015, [10.1109/GLOCOM.2015.7417118](https://doi.org/10.1109/GLOCOM.2015.7417118).
23. Fazeldehkordi, E., Owe, O., Noll, J., *Security and Privacy in IoT Systems: A Case Study of Healthcare Products*, 13th International Symposium on Medical Information and Communication Technology (ISMICT).

HUMANITIES ELECTIVE COURSES

IHSXXX Introduction to Sociology [3-0-0-3]

Course Objectives

- To understand the relationship between Science, Technology, Society, and Culture.
- To familiarize with the social processes and dynamics of scientific knowledge and developments in Social Sciences.
- To introduce Sociology as a science of society vis-à-vis Science and Technology.
- To know the fundamental social processes and institutions such as socialization, marriage, family, caste, and gender.
- To discuss some of the important issues and problems in India today.

Course Outcomes

Students who successfully complete this course will be able to:

- Understand the relationship between Science, Technology, Society, and Culture.
- Analyze social processes and dynamics of scientific knowledge and developments in Social Sciences.
- Explain Sociology as a science of society vis-à-vis Science and Technology.
- Identify the fundamental social processes and institutions of socialization, marriage, family, caste, and gender.
- Discuss important issues and problems in India today, such as Communalism, Diversity and Pluralism, Fundamentalism, Secularism, Globalization, Liberalization, and the Information Communication Revolution.

Syllabus

Sociology and Social Sciences: Science, Social Science, and Society, Scientific Study of Society, Sociology as the Science of Society; Approaches and Methods: Macro and Micro – Object and Subject, Objectivist and Interpretivist.

Society, Social Structure, and Culture: Nature and Culture: Science and Society Interface, Society and Social Structure, Culture, Socialization, Social Institutions – Marriage, Family, Caste, Class, and Gender.

India Today – Major Issues and Challenges: Diversity and Pluralism, Secularism and Communalism, Globalization and Consumer Culture, Information Communication Revolution – New Media.

Textbooks/References

1. Haralambos, Michael with Robin Heald, *Sociology – Themes and Perspectives*, Oxford University Press, New Delhi, 1980.

2. Mohanty, Manoranjan, *Class, Caste, Gender*, Sage Publications, New Delhi, 2004.
3. Perry, John, and Perry, Erna, *Contemporary Society: An Introduction to Social Science*, Allyn & Bacon, London, 2008.
4. Bryman, Alan, *Social Research Methods*, Oxford University Press, Oxford, 2005.
5. Srinivas, M.N., *Caste in India and Other Essays*, Asia Publishing House, Delhi, 1962.
6. Patel, Tulasi, *The Family in India: Structure and Practice*, Sage Publications, New Delhi, 2005.
7. Knorr-Cetina, Karin and Michael Mulkay, *Science Observed: Perspectives on the Social Study of Science*, Sage Publications, London, 1983.
8. Merton, Robert K., *The Sociology of Science*, University of Chicago Press, Chicago, 1973.
9. Bijker, W.E., *Social Construction of Technology*, International Encyclopedia of the Social and Behavioural Sciences, edited by N. Smelser and P. Bailes, Amsterdam, 2001.
10. Franklin, Sarah, *Science as Culture, Cultures of Science*, Annual Review of Anthropology, 1995.
11. Sharma, K.L., *Indian Social Structure and Change*, Rawat Publications, Jaipur, 2007.
12. Latour, Bruno, and Steve Woolgar, *Laboratory Life: The Construction of Scientific Facts*, Princeton University Press, Princeton NJ, 1986.

IHSXXX Taxation and Human Development / Economics [3-0-0-3]

Course Objectives

The course seeks to sensitize the students to concepts of income, human development, government's role in development, governance functions, and taxation. It aims at raising the capacity of non-economics students to appreciate the need to understand some of these concepts for good citizenship as well as for becoming able technologists.

Syllabus

Taxation and Central Transfers: Introduction to National Income, Gross Domestic Product (GDP), Gross National Income (GNI), Three Approaches to GDP, System of National Accounts and Production Boundary.

Household's Non-Market Production: Women, Work, and the Economy: Macroeconomic Gains from Gender Equity, Measuring Government Output, Satellite Account, Tourism Satellite Account for India.

GDP to Welfare Measures: Human Development Index, Purchasing Power Parities, National Income, and Quarterly

Estimates of Gross Domestic Product (mospi.nic.in).

Textbooks/References

1. Ann Chadeau, *What Is Households' Non-Market Production Worth?*, OECD Economic Studies No. 18, Spring 1992.
2. Katrin Elborgh-Woytek et al., *Women, Work, and the Economy: Macroeconomic Gains from Gender Equity*, IMF SDN/13/10, 2013.
3. Lequiller, F. and D. Blades, *Understanding National Accounts: Second Edition*, OECD Publishing, 2014. (<http://dx.doi.org/10.1787/9789264214637-en>)
4. UNDP, *Human Development Report*, 1990 and 2016.

IHSXXX Introduction to Digital Humanities [1-0-0-1]

Course Description

Digital Humanities (DH) looks at how the study of humanities has been immensely transformed by digital technology. It is an emerging discipline that studies the changing methodologies and pedagogies in humanities study and research in the light of the growing intersection of humanities with digital technology. This approach has a new relevance in the context of the COVID-19 pandemic, which has delivered a knockout blow to traditional approaches in education.

Digital Humanities has a two-way approach: the systematic use of digital resources in humanities and the humanistic analysis of the application of digital resources. The discipline requires a certain amount of acquaintance with computer technology, and the course can be handled in collaboration with faculties from the Computer Science department.

Course Objectives

- To introduce the basic concepts and tools in Digital Humanities.
- To familiarize students with new forms of textuality and digital archiving.
- To develop competency in digital writing.
- To incorporate digital tools and methods to support their own core studies and to develop practical skills in electronic archiving, processing, editing, etc.

Course Outcomes

By the end of the course, students will be able to:

- Critically discuss digital humanities projects in light of current methods and theoretical approaches to the field.
- Apply principles of textual, editorial, and communication theory to technical situations, enabling them to produce high-quality work in a digital environment.

Syllabus

A history of the Humanities in an age of digital computing – Transitions in humanities – Development of digital humanities –

Beginnings of the intersection of humanities with computing – Ethics and the digital community – Digital Libraries and Archiving – Digital Humanities in India – Significant Digital Humanities Projects – Intersections of Digital Humanities with web designing, software programming, technical writing, Game Studies, etc. – Job prospects: Data Manager, Data Designer, Digital Curator, Cultural Designer, Information Architect, Data Scientist, Metadata Analyst – Related disciplines (Medical Humanities, Astrobiology, etc.).

Textbooks/References

1. Maya Dodd and Nidhi Kalra (Eds.), *Exploring Digital Humanities in India: Pedagogies, Practices, and Institutional Possibilities*, Routledge India, 2020.
2. Schreibman, Siemens, and Unsworth (Eds.), *A Companion to Digital Humanities*, Blackwell, 2004.
3. Matthew Gold (Ed.), *Debates in the Digital Humanities*, University of Minnesota Press, 2012.
4. Peter Shillingsburg, *From Gutenberg to Google: Electronic Representations of Literary Texts*, Cambridge University Press, 2006.

IHSXXX Visual Communication [1-0-0-1]

Course Description

Visual Communication examines the production, reception, and interpretation of the aspects of culture that communicate through visual imagery. These include not only art but also photography and commercial imagery (including advertisements, popular film, and television). The emergence of visual culture is closely linked to the development of visual technologies.

Course Objectives

- To understand the historical, physiological, psychological, perceptual, and cognitive aspects of visual communication.
- To critically analyse the ethical implications of electronic and print media images.
- To examine how visual culture is inextricably tied to marketing and consumerism in the modern world.
- To explore the possibilities of digital marketing.

Course Outcomes

By the end of the course, students will be able to:

- Build a critical understanding of the relationship between images and their representation.
- Express their own ideas with creativity, clarity, and critical thinking.

Syllabus

Introduction to Visual Communication – The politics of visibility – The virtual ‘real’ in contemporary culture – Consumption of mass culture – Photography, Cinema, Advertisement, Web content, etc. – The Digital World: The Virtual Reality in Contemporary Culture – Theories of Visuality and impact on visual communication – New perspectives in visual culture – Virtual

usurps in the internet and cyberspaces – The misuse of visual technologies.

Textbooks/References

1. Lester, Paul Martin. *Visual Communication: Images with Messages* (4th Edition). Belmont, CA: Wadsworth Publishing Company, 2006.
2. Jones, Steve. *Virtual Culture: Identity and Communication in Cybersociety*. New Delhi: Sage, 2002.

IHSXXX Science Fiction and Graphic Narratives [1-0-0-1]

Course Description

The course looks at the origins of science fiction and graphic narratives to understand how science and technology intersect with the world of fiction. This course is designed for a reader community who will approach these works from a critical perspective.

Course Objectives

- To familiarize students with the history, culture, and origins of science fiction and graphic narratives.
- To appreciate the technological elements in science fiction and graphic narratives.
- To promote the close reading of science fiction and visual storytelling materials.

Course Outcomes

By the end of the course, students will be able to:

- Identify the relationship between science and society.
- Write analytically about fictional works.

Syllabus

Origins of science fiction – Sub-genres and themes – Utopia – Dystopia – Time travel – Cybernetics – Issues of gender, ecology, etc., in science fiction. Origins of graphic narratives – Recent trends – Emergence of web comics – Print versus web comics – Comics and technology.

Textbooks/References

1. Paul Gravett. *Graphic Novels*. New York: Collins, 2005.
2. Bould, Mark et al. (Eds.), *The Routledge Companion to Science Fiction*. Routledge, London & New York, 2009.

IHSXXX Professional Communication [1-0-0-1]

Course Description

The course covers the important aspects of communication in the professional scenario. Emphasis is given to key language skills essential for a successful professional life.

Course Objectives

- To understand the importance of communication in the professional/corporate world.
- To attain fluency in oral communication.

- To attain proficiency in technical writing.
- To develop positive body language.
- To learn the etiquettes to be practiced while participating in seminars and group discussions.
- To acquire interview skills.
- To practice interpersonal skills for better relations with seniors, juniors, peers, and stakeholders.

Course Outcomes

By the end of the course, students will be able to:

- Draft effective proposals, circulars, minutes, etc.
- Prepare an attractive resume.
- Apply for a job and participate in the selection process.
- Overcome the fear of interviews and be ready to perform at different levels of the selection process.
- Project a good personal image and social etiquette to have a positive impact on building their chosen career.

Syllabus

Basics of professional/business communication – Developing social communication skills – Vocabulary in the professional scenario – Relevance of non-verbal communication – Body language – Haptics – Chronemics. Technical writing – Writing proposals – Email etiquettes – Basics of research writing – Abstract preparation – Drafting a resume – Cover letter – Digital profile – Video profile. Interview skills – Group discussion.

Textbooks/References

1. Tebeaux, Elizabeth, and Sam Dragga. *The Essentials of Technical Communication*. OUP, 2018.
2. Raman, Meenakshi, and Sangeeta Sharma. *Technical Communication: Principles and Practice*. OUP, 2015.

IHSXXX Advanced Communication Skills [1-0-0-1] (Non-Credit Course)

Course Objectives

This course is designed to build the vocabulary and language proficiency that are appropriate for IT professionals. This course will enable them to recognize problems with listening style and help them use listening skills to meet various professional needs. Effective communication abilities are helpful in comprehending, developing, and presenting technical presentations. Active listening is necessary to grasp the context in order to solve problems. To eliminate uncertainty during team activities, writing abilities must be standardized and sound.

Course Outcomes

After successful completion of this course, students will be able to:

- Understand and use various listening and speaking etiquette in everyday communication.
- Successfully read, comprehend, and critically evaluate both technical and non-technical materials.

- Adapt writing techniques that are successful for both non-technical and technical communication.

Syllabus

Listening: Active and passive listening, intensive listening, developing effective listening skills, barriers to effective listening, listening to longer technical talks, listening to classroom lectures, listening to documentaries and making notes, TED talks. Exercises based on audio materials like radio and podcasts, listening to songs, practice and exercises.

Spoken: Phonetics, pronunciation practice: problem sounds, MMFS (Multimedia Feedback System), stress, intonation, pitch, uses of a dictionary for pronunciation practice. Conversation practices.

Reading, comprehension, and summarizing: Reading styles, speed, valuation, critical reading, SQ3R method, PQRS method – Speed reading – Comprehension: techniques, understanding textbooks, marking and underlining – Note-taking: recognizing non-verbal cues.

Formal writing: Technical writing: differences between technical and literary style – Letter writing (formal, informal, and semi-formal), job applications, minute preparation, CV preparation (differences between Bio-Data, CV, and Resume), and reports – Elements of style – Common errors in writing, describing a process, Statement of Purpose, instructions, checklists.

Textbooks/References

1. John Seely, *The Oxford Guide to Writing and Speaking*, Oxford University Press, New Delhi, 2004.
2. Turton, N.D and Heaton, J.B, *Dictionary of Common Errors*, Addison Wesley Longman Ltd., Indian reprint 1998.
3. Anderson, P.V, *Technical Communication*, Thomson Wadsworth, Sixth Edition, New Delhi, 2007.

IHSXXX Soft Skills and Professional Ethics

Course Objectives

This course is designed to improve students' soft skills and make them aware of work ethics. Students should be able to take part in debates, group discussions, interviews, and presentations, making use of effective communication and presentation techniques. Since good communication skills help further one's career, every effort will be made to increase the students' awareness of appropriate professional techniques. It is also designed to develop their writing and communication skills. This course will give more focus to the practical application of each module.

Course Outcomes

After successful completion of this course, students will be able to:

- Develop soft skills.
- Groom corporate habits.
- Face interviews with confidence.
- Be aware of work ethics.
- Create the right impression in interviews.

- Receive guidance for paper presentations.

Syllabus

Soft Skills: SMART Goals, Teamwork, Autodidacticism, Time Management, Stress Management, Positive Attitude, SWOT Analysis.

Interview Skills: Self-Awareness, Confidence, Resume Writing, Body Language, Interview FAQs, Handling Rejections.

Group Discussions & Mock Interviews: Practical exercises.

Writing Skills: Article writing, Paper presentation for IT students, Analytical and issue-based essays, Report writing: basics of report writing, structure of a report, types of reports, references, bibliography – Referencing style (IEEE Format) – Plagiarism.

Professional Ethics: Responsibilities and Rights of Professionals, Managing Multiple Roles, Public Relations.

Modern-day Research and Study Skills: Search engines and how to get optimal results using language, repositories,

forums such as GitHub, Stack Exchange, OSS communities (MOOC, SWAYAM, NPTEL), and Quora.

Textbooks/References

1. Stephen E. Lucas, *The Art of Public Speaking*, 10th Edition, McGraw Hill Education, 2012.
2. David F. Beer and David McMurrey, *Guide to Writing as an Engineer*, John Wiley, New York, 2004.
3. Goodheart-Willcox, *Professional Communication*, First Edition, 2017.
4. Gopalaswamy Ramesh and Mahadevan Ramesh, *The Ace of Soft Skills: Attitude, Communication and Etiquette for Success*, Pearson Education, 1st edition, 2013.
5. Samson et al., *English for Life-4*, New Delhi: Cambridge UP.